

PROBABILITY AND STATISTICS

Probabilities arise in many problems dealing with random events or large numbers of particles defining random variables. An event is called **random** if it is practically impossible to predict from the initial state. This includes cases where we have merely incomplete information about initial states and/or the dynamics. For example, in statistical mechanics we deal with systems containing large numbers of particles, but our knowledge is ordinarily limited to a few average or macroscopic quantities such as the total energy, the volume, the pressure, or the temperature. Because the values of these macroscopic variables are consistent with very large numbers of different microscopic configurations of our system, we are prevented from predicting the behavior of individual atoms or molecules. Often the average properties of many similar events are predictable, as in quantum theory. This is why probability theory can be and has been developed.

Random variables are also involved when data depend on chance, such as weather reports and stock prices. The theory of probability describes mathematical models of chance processes in terms of probability distributions of random variables that describe how some “random events” are more likely than others. In this sense, probability is a measure of our ignorance, giving quantitative meaning to qualitative statements, such as “It will probably rain tomorrow” and “I’m unlikely to draw the queen of hearts.” Probabilities are of fundamental importance in quantum mechanics and statistical mechanics and are applied in meteorology, economics, games, and many other areas of daily life.

Because experiments in the sciences are always subject to measurement errors, theories of errors and their propagation involve probabilities. **Statistics** is the area of mathematics that connects observations on data samples to inferences about the probable content of the entire population from which the sample(s) came. It is an extensive and sophisticated branch of mathematics, and in the present text only a few of the most basic concepts can be presented. The material found here may be adequate to provide a conceptual basis for statistical mechanics, but can at best be an elementary introduction to the ideas needed to

gain maximum information from data-intensive experimental studies such as those arising from the study of cosmic rays or the data from high-energy particle accelerators. A more complete picture of the role of statistics in physics and engineering can be obtained from a number of the texts in the Additional Readings.

23.1 PROBABILITY: DEFINITIONS, SIMPLE PROPERTIES

All possible **mutually exclusive**¹ outcomes of an experiment that is subject to chance represent the events (or points) of a **sample space** S . Suppose we toss a coin, and record that it lands either “heads” or “tails.” These are mutually exclusive events, so our sample space for a single coin toss can be deemed to be spanned by a discrete random variable x , with possible values x_i , which (based on our experiment, called a **trial**), will have one of the two values x_1 (for heads) or x_2 (for tails). Now suppose, with the same sample space, we carry out larger numbers of trials. Some will have the result x_1 (heads), others x_2 (tails). It is of interest to define the **probability** of an outcome in our sample space by the ratio

$$P(x_i) \equiv \frac{\text{number of times event } x_i \text{ occurs}}{\text{total number of trials}}, \quad (23.1)$$

where it is assumed that the number of trials is large enough that $P(x_i)$ approaches a constant limiting value. In the event that we are able to enumerate all the possible events that produce outcomes in our sample space and can also assume that each event is equally likely, we may then define the theoretical probability of an outcome x_i as

$$P(x_i) \equiv \frac{\text{number of outcomes } x_i}{\text{total number of all events}}. \quad (23.2)$$

An example of the use of this theoretical probability can be illustrated using coin tosses. For example, suppose that we toss a coin twice and take our random variable x to be the number of heads obtained in a two-toss trial. Our sample S now contains three possible values of x , which we designate x_0 , x_1 , x_2 , where we are now letting x_i stand for the occurrence of i heads in the two tosses. Obviously, the only possible values of x are 0, 1, and 2. But we also know that the four possible results of two successive tosses are (heads, then heads), (heads, then tails), (tails, then heads), (tails, then tails); these possibilities are mutually exclusive and it is reasonable to assume that they are equally likely. Then, using [Eq. \(23.2\)](#), we conclude that the probabilities of x_2 (two heads) and x_0 (no heads) will each be $1/4$, while the probability of x_1 (one heads) will be $1/2$.

The experimental definition, [Eq. \(23.1\)](#), is the more appropriate when the total number of events is not well defined (or is difficult to obtain) or we cannot identify equally likely outcomes. A large, thoroughly mixed pile of black and white sand grains of the same size and in equal proportions is a relevant example, because it is impractical to count them all. But we can count the grains in a small sample volume that we pick. This way we can check that white and black grains turn up with roughly equal probability $1/2$, provided that we put back each sample and mix the pile again. It is found that the larger the sample volume,

¹This means that given that one particular event did occur, the others could not have occurred.

the smaller the spread in probability about $1/2$ will be. Moreover, the more trials we run, the closer the average of all the individual trial probabilities will be to $1/2$. We could even pick single grains and check if the probability $1/4$ of picking two black grains in a row equals that of two white grains, etc. There are lots of statistics questions we can pursue. Thus, piles of colored sand provide for instructive experiments.

The following axioms are self-evident.

- Probabilities satisfy $0 \leq P \leq 1$. Probability 1 means certainty; probability 0 means impossibility.
- The entire sample has probability 1. For example, drawing an arbitrary card from a deck of cards has probability 1.
- The probabilities for mutually exclusive events add. The probability for getting exactly one head in two coin tosses is $1/4 + 1/4 = 1/2$ because it is $1/4$ for head first and then tail, plus $1/4$ for tail first and then head.

Example 23.1.1 PROBABILITY FOR A OR B

Before proceeding with this example, we must clarify the definition of “or.” In probability theory, “ A or B ” means A , B , or **both** A and B . The specification “ A or B but not both” is referred to as the **exclusive or** of A and B (sometimes abbreviated **xor**).

What is the probability for drawing a club or a jack from a shuffled deck of cards?² To answer this question we need to identify equally probable mutually exclusive events. We note that because there are 52 cards in a deck, the drawing of each being equally likely (with 13 cards for each suit and 4 jacks), there are 13 clubs including the club jack, and 3 other jacks; that is, there are 16 mutually exclusive draws that meet our specification out of the total of 52, giving the probability $(13 + 3)/52 = 16/52 = 4/13$. ■

Sets, Unions, and Intersections

If we represent a sample space by a set S of points, then events meeting certain specifications can be identified as subsets $A, B, \dots$ of S , denoted as $A \subset S$, etc. Two sets A, B are equal if A is contained in B , denoted $A \subset B$, and B is contained in A , denoted $B \subset A$. The **union** $A \cup B$ consists of all points (events) that are in A or B or both (see Fig. 23.1). The **intersection** $A \cap B$ consists of all points that are in both A and B . If A and B have no common points, their intersection is the **empty set** (which has no elements), and we write $A \cap B = \emptyset$. The set of points in A that are not in the intersection of A and B is denoted by $A - A \cap B$, thereby **defining a subtraction of sets**. If we take the club suit in Example 23.1.1 as set A and the four jacks as set B , then their union comprises all clubs and jacks, and their intersection is the club jack only.

Each subset A has its probability $P(A) \geq 0$. In terms of these set-theory concepts and notations, the probability laws we just discussed become

$$0 \leq P(A) \leq 1.$$

²Note that these events are not mutually exclusive.

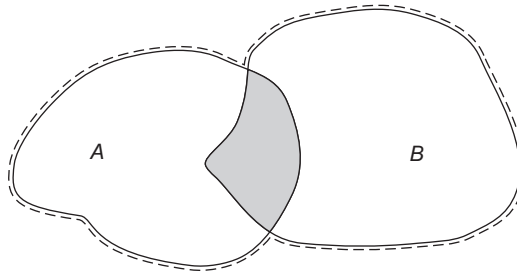


FIGURE 23.1 The shaded area gives the intersection $A \cap B$, corresponding to the A and B event sets; the dashed line encloses $A \cup B$, corresponding to the event set A or B .

The entire sample space has $P(S) = 1$. The probability of the union $A \cup B$ of mutually exclusive events is the sum

$$P(A \cup B) = P(A) + P(B), \quad \text{where } A \cap B = \emptyset. \quad (23.3)$$

The **addition rule** for probabilities of arbitrary sets is given by the following theorem:

$$\text{Addition rule: } P(A \cup B) = P(A) + P(B) - P(A \cap B). \quad (23.4)$$

To prove Eq. (23.4), we write the union as two mutually exclusive sets: $A \cup B = A \cup (B - B \cap A)$, where we have subtracted the intersection of A and B from B before joining them. The respective probabilities of these mutually exclusive sets are $P(A)$ and $P(B) - P(B \cap A)$, which we add. We could also have written $A \cup B = (A - A \cap B) \cup B$, from which our theorem follows similarly by adding these probabilities: $P(A \cup B) = [P(A) - P(A \cap B)] + P(B)$. Note that $A \cap B = B \cap A$. The relationships among these sets can be checked by referring to Fig. 23.1.

Sometimes the rules and definitions of probabilities that we have discussed so far are not sufficient, and we need to introduce the notion of conditional probability. Let A and B denote sets of events in our sample space. The **conditional probability** $P(B|A)$ is defined to be the probability that an event which is a member of A is also a member of B . To understand the need for this somewhat formal definition, consider the following example.

Example 23.1.2 CONDITIONAL PROBABILITY

Consider a box of 10 identical red and 20 identical blue pens, from which we remove pens successively in a random order without putting them back. Suppose we draw a red pen first, event R , followed by the draw of a blue pen, event B . One way to compute $P(R, B)$ is to note that our sample space consists of 30×29 mutually exclusive and equally probable points (each a two-event ordered sequence), of which 10×20 meet our specifications, leading to the computation $P(R, B) = (10 \times 20)/(30 \times 29) = 20/87$. Note that in this example, $P(R, B)$ refers to **ordered** events.

Another way of making the same computation is to start by noting that the initial drawing of a red pen will occur with probability $P(R) = 10/30$. But now the probability of drawing a blue pen in the next round, event B , however, will depend on the fact that we drew a red pen in the first round, and is given by the **conditional probability** $P(B|R)$. Since there are

now 29 pens of which 20 are blue, we easily compute $P(B|R) = 20/29$, and the probability of the sequence “red, then blue” is

$$P(R, B) = \frac{10}{30} \frac{20}{29} = \frac{20}{87}, \quad (23.5)$$

equal to the result we obtained previously. ■

The generalization of the result in Eq. (23.5) is the very useful formula

$$P(A, B) = P(A)P(B|A), \quad (23.6)$$

which has the obvious interpretation that the probability that A and B both occur can be written as the probability of A , multiplied by the conditional probability $P(B|A)$ that B occurs, given the occurrence of A .

Two observations relative of Eq. (23.6) are in order. First, it can be rearranged to reach an explicit formula for $P(B|A)$:

$$P(B|A) = \frac{P(A, B)}{P(A)}. \quad (23.7)$$

Second, if the conditional probability $P(B|A) = P(B)$ is independent of A , then the events A and B are called **independent**, and the combined probability is simply the **product of both probabilities**, or

$$P(A, B) = P(A)P(B), \quad (A \text{ and } B \text{ independent}). \quad (23.8)$$

If A and B are defined in a way that neither depends on the other (a condition not satisfied in Example 23.1.2), we can rewrite Eq. (23.7) as

$$P(B|A) = \frac{P(A \cap B)}{P(A)}. \quad (23.9)$$

Example 23.1.3 SCHOLASTIC APTITUDE TESTS

Colleges and universities rely on the verbal and mathematics SAT scores, among others, as predictors of a student's success in passing courses and graduating. A research university is known to admit mostly students with a combined verbal and mathematics score of 1400 points or more. The graduation rate is 95%; that is, 5% drop out or transfer elsewhere. Of those who graduate, 97% have an SAT score of at least 1400 points, while 80% of those who drop out have an SAT score below 1400. Suppose a student has an SAT score below 1400. What is his/her probability of graduating?

Let A represent all students with an SAT test score below 1400, and let B represent those with scores ≥ 1400 . These are mutually exclusive events with $P(A) + P(B) = 1$. Let C represent those students who graduate, and let $\bar{C}$ represent those who do not. Our problem here is to determine the conditional probabilities $P(C|A)$ and $P(C|B)$. To apply Eq. (23.9) we need the four probabilities $P(A)$, $P(B)$, $P(A \cap C)$, and $P(B \cap C)$.

Among the 95% of students who graduate, 3% are in set A and 97% are in set B , so

$$P(A \cap C) = (0.95)(0.03) = 0.0285, \quad P(B \cap C) = (0.95)(0.97) = 0.9215.$$

Among the 5% of students who do not graduate, 80% are in set A and 20% are in set B , so

$$P(A \cap \tilde{C}) = (0.05)(0.80) = 0.0400, \quad P(B \cap \tilde{C}) = (0.05)(0.20) = 0.0100.$$

Since $P(A) = P(A \cap C) + P(A \cap \tilde{C})$, and likewise for $P(B)$, we have

$$P(A) = 0.0285 + 0.0400 = 0.0685, \quad P(B) = 0.9215 + 0.0100 = 0.9315.$$

Now, applying Eq. (23.9), we obtain the final results

$$P(C|A) = \frac{P(A \cap C)}{P(A)} = \frac{0.0285}{0.0685} \approx 41.6\%,$$

$$P(C|B) = \frac{P(B \cap C)}{P(B)} = \frac{0.9215}{0.9315} \approx 98.9\%;$$

that is, a little less than 42% is the probability for a student with a score below 1400 to graduate at this particular university. ■

As a corollary to the equation for conditional probability, Eq. (23.9), we now compare $P(A|B) = P(A \cap B)/P(B)$ and $P(B|A) = P(A \cap B)/P(A)$, obtaining a result known as **Bayes' theorem**:

$$P(A|B) = \frac{P(A)}{P(B)} P(B|A). \quad (23.10)$$

Bayes' theorem is a special case of the following more general theorem:

If the random events A_i with probabilities $P(A_i) > 0$ are mutually exclusive and their union represents the entire sample S , then an arbitrary random event $B \subset S$ has the probability

$$P(B) = \sum_{i=1}^n P(A_i) P(B|A_i). \quad (23.11)$$

The decomposition law given by Eq. (23.11) resembles the expansion of a vector into a basis of unit vectors defining its components. This relation follows from the obvious decomposition $B = \cup_i (B \cap A_i)$ (this notation indicates the union of all the quantities $B \cap A_i$, see Fig. 23.2), which implies $P(B) = \sum_i P(B \cap A_i)$ because the components $B \cap A_i$ are mutually exclusive. For each i , we know from Eq. (23.9) that $P(B \cap A_i) = P(A_i) P(B|A_i)$, which proves the theorem.

Counting Permutations and Combinations

Counting the events in samples can help us find probabilities; this procedure is found to be of great importance in statistical mechanics.

If we have n different molecules, let us ask in how many ways we can arrange them in a row, that is, permute them. This number is defined as the number of their **permutations**. Thus, by definition, the **order matters in permutations**. There are n choices of picking the first molecule, $n - 1$ for the second, etc. Altogether there are $n!$ permutations of n **different** molecules or objects.

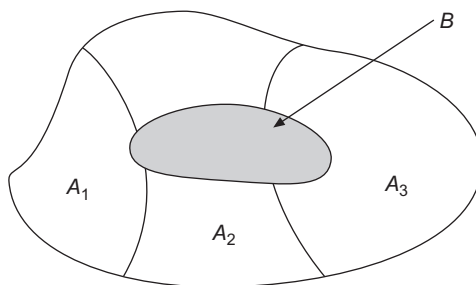


FIGURE 23.2 The shaded area B is composed of mutually exclusive subsets of B belonging also to A_1, A_2, A_3 , where the A_i are mutually exclusive.

Generalizing this, suppose there are n people but only $k < n$ chairs to seat them. In how many ways can we seat k people in the chairs? Counting as before, we get

$$n(n-1) \cdots (n-k+1) = \frac{n!}{(n-k)!} \quad (23.12)$$

for the number of permutations of k objects which can be formed by selection from a set originally containing n objects.

We now consider the counting of **combinations** of objects, where the term **combination** is defined to refer to sets in which the object order is irrelevant. For example, three letters a, b, c can be combined, two letters at a time, in three ways: ab, ac, bc . If letters can be repeated, then we also have the pairs aa, bb, cc and have a total of six combinations. These examples illustrate the fact that a **combination** of different particles differs from a **permutation** in that the particles' order **does not matter**. Combinations may occur with repetition or without; the essential point is that no two combinations contain the same particles.

The number of different combinations of n numbered (and thereby distinguishable) particles, k at a time and without repetitions, is given by the binomial coefficient

$$\frac{n(n-1) \cdots (n-k+1)}{k!} = \binom{n}{k}. \quad (23.13)$$

To prove Eq. (23.13), we start from the number $n!/(n-k)!$ of permutations in which k particles were chosen from n , and divide out the number $k!$ of permutations of the group of k particles because their order does not matter in a combination.

A generalization of the above is a situation in which we have a total of n distinguishable (numbered) objects, and we place n_1 of these into Box 1, n_2 into Box 2, etc. We wish to know how many different ways this can be done (this is a **combination** problem because the objects in each box do not form ordered sets). A simple way to solve this problem is to identify each permutation of the n objects with an assignment into boxes; the first n_1 of the permuted objects is placed in Box 1, the next n_2 in Box 2, etc. However, permutations that differ only in the ordering of objects destined for the same box do not constitute different distributions, so the total number of distributions will be $n!$ (the overall number

of permutations) divided by $n_1!, n_2!, \text{etc.}$ Thus, our overall formula is

$$B(n_1, n_2, \dots) = \frac{n!}{n_1! n_2! \dots}. \quad (23.14)$$

This quantity is sometimes referred to as a **multinomial coefficient**; if there were only two boxes it reduces to the binomial coefficient.

For a related problem with repetition, suppose that we have an unlimited supply of particles bearing each number from 1 through k . Then the number of distinct ways in which n particles can be chosen can be shown to be

$$\binom{n+k-1}{n} = \binom{n+k-1}{k-1}. \quad (23.15)$$

The following example provides a proof of Eq. (23.15).

Example 23.1.4 COMBINATIONS WITH REPETITION

The physical relevance of the situation giving rise to Eq. (23.15) is that it is mathematically equivalent to the number of ways that n identical, indistinguishable particles can be placed in k boxes. To see that these problems are equivalent, note that the number on each particle of Eq. (23.15) can be used to identify the box in which that particle will be placed.

A simple way to count the possible assignments is to consider the distinguishable ways that n indistinguishable particles and $k-1$ indistinguishable partitions can be placed in a line containing $n+k-1$ items. The particles (if any) that occur in the line earlier than the first partition are assigned to Box 1; those between the first and second partitions are assigned to Box 2, etc., with the particles (if any) occurring later than the $(k-1)$ th (the last) partition are assigned to Box k . Each different placement of the partitions yields a unique assignment of particles to boxes, and the number of different partition placements is the number of combinations given by the binomial coefficient in Eq. (23.15). ■

In statistical mechanics, we frequently need to know the number of ways in which it is possible to put n particles in k boxes subject to various additional specifications. If we are working in classical theory, our more complete specification includes the notion that the particles are **distinguishable**, and we refer to the probability computation as that given by **Maxwell-Boltzmann statistics**. In the quantum domain, it is assumed that identical particles are inherently indistinguishable; in fact, we cannot even identify them by their trajectories, as the notion of path is blurred by the Heisenberg uncertainty principle. This indistinguishability leads to the requirement that many-particle states must have symmetry under the interchange of identical particles, and in nature we find two cases: Either the wave function is symmetric under interchange of the coordinates of a pair of identical particles (such particles are said to exhibit **Bose-Einstein statistics**), or the coordinate interchange causes a reversal in the sign of the wave function (the case called **Fermi-Dirac statistics**). The symmetry (or antisymmetry) under particle interchange influences the way in which particles can be assigned to states (boxes): In Bose-Einstein statistics

any number of **indistinguishable** particles may be placed in the same box; in Fermi-Dirac statistics no box may contain more than one **indistinguishable** particle.

Application of the various kinds of statistics in general problems is outside the scope of this text; however, the basic case in which we simply count the number of assignments that are possible is easily approached. If we have n particles and k available states:

- In Maxwell-Boltzmann (classical) statistics, the number of possible assignments of particles to states is k^n (each particle can independently be assigned to any state).
- In Bose-Einstein statistics, the number of possible assignments is given by Eq. (23.15).
- In Fermi-Dirac statistics, the number of possible assignments is $\binom{k}{n}$. This formula gives the number of ways that n of the k states can be selected for occupancy. Note that the number of assignments is zero if $n > k$, indicating that we cannot make any assignment (with a maximum of one particle per state) unless there are at least as many states as there are particles.

Exercises

- 23.1.1** A card is drawn from a shuffled deck. (a) What is the probability that it is black, (b) a red nine, (c) or a queen of spades?
- 23.1.2** Find the probability of drawing two kings from a shuffled deck of cards (a) if the first card is put back before the second is drawn, and (b) if the first card is not put back after being drawn.
- 23.1.3** When two fair dice are thrown, what is the probability of (a) observing a number less than 4, or (b) a number greater than or equal to 4 but less than 6?
- 23.1.4** Rolling three fair dice, what is the probability of obtaining six points?
- 23.1.5** Determine the probability $P(A \cap B \cap C)$ in terms of $P(A)$, $P(B)$, $P(C)$, $P(A \cup B)$, $P(A \cup C)$, $P(B \cup C)$, and $P(A \cup B \cup C)$.
- 23.1.6** Determine directly or by mathematical induction (Section 1.4) the probability of a distribution of N (Maxwell-Boltzmann) particles in k boxes with N_1 in Box 1, N_2 in Box 2, ..., N_k in the k th box for any numbers $N_j \geq 1$ with $N_1 + N_2 + \cdots + N_k = N$, $k < N$. Repeat this for Fermi-Dirac and Bose-Einstein particles.
- 23.1.7** Show that $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$.
- 23.1.8** Determine the probability that a positive integer $n \leq 100$ is divisible by a prime number $p \leq 100$. Verify your result for $p = 3, 5, 7$.
- 23.1.9** Put two particles obeying Maxwell-Boltzmann (Fermi-Dirac, or Bose-Einstein) statistics in three boxes. How many ways of doing so are there in each case?

23.2 RANDOM VARIABLES

In this section we define properties that characterize the probability distributions of random variables, by which we mean variables that will assume various numerical values with individual probabilities. Thus, the name of a color (e.g., “black” or “white”) cannot be the value assigned a random variable, but we can define a random variable to have one numerical value for “black” and another for “white”; the usefulness of our definition may depend on the problem we are attempting to solve.

Having defined a random variable and given its distribution, we are interested in particular in its **mean** or **average** value, and in measures of the width or spread of its values. The width is of particular importance when the random variable represents repeated measurements of the same quantity but subject to experimental error. In addition, we introduce properties that characterize the extent to which the value of one random variable depends on (i.e., is **correlated** with) those of another.

Random variables can be **discrete**, as for example those introduced in the previous section to describe the outcomes of coin tosses, or they may be **continuous**, either inherently so (as, for example, the wave function in a quantum mechanical system) or because they consist of so many closely spaced discrete points that it is impractical to work with them individually.

Example 23.2.1 DISCRETE RANDOM VARIABLE

The possible outcomes of the tossing of a die define a random variable X with values $x_1, x_2, \dots, x_6$, each with probability $1/6$; we can denote this by writing $P(x_i) = 1/6, i = 1 \dots 6$.

If we toss two dice and record the sum of the points shown in each trial, then this sum is also a discrete random variable, which takes on the value 2 when both dice show 1 with probability $(1/6)^2$; the value 3 in either of the two cases in which one die has 1 and the other 2, hence with probability $(1/6)^2 + (1/6)^2 = 1/18$. Continuing, the value 4 is reached in three equally probable ways: $2 + 2, 3 + 1$, and $1 + 3$ with total probability $3(1/6)^2 = 1/12$; the values 5 and 6 are reached with the respective probabilities $4(1/6)^2 = 1/9$ and $5(1/6)^2 = 5/36$; and the value 7 occurs with the maximum probability, $6(1/6)^2 = 1/6$. The value 8 is reached in five ways ($6 + 2, 5 + 3, 4 + 4, 3 + 5, 2 + 6$), with probability $5(1/6)^2 = 5/36$, and further increases in x lead to smaller probabilities, finally at $x = 12$ reaching probability $(1/6)^2 = 1/36$. This probability distribution is symmetric about $x = 7$, and can be represented graphically as in Fig. 23.3 or algebraically as

$$P(x) = \frac{x-1}{36} = \frac{6-(7-x)}{36}, \quad x = 2, 3, \dots, 7,$$

$$P(x) = \frac{13-x}{36} = \frac{6+(7-x)}{36}, \quad x = 7, 8, \dots, 12.$$



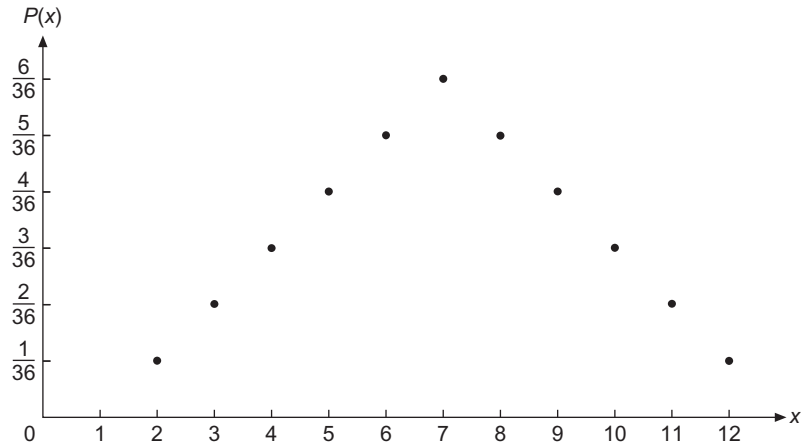


FIGURE 23.3 Probability distribution $P(x)$ of the sum of points when two dice are tossed.

In summary, then,

- If a discrete random variable X can assume the values x_i , each value occurs by chance with a probability $P(X = x_i) = p_i \geq 0$ that is a discrete-valued function of the random variable X , and the probabilities satisfy $\sum_i p_i = 1$.
- We define the probability density $f(x)$ of a **continuous random variable** X as

$$P(x \leq X \leq x + dx) = f(x)dx; \quad (23.16)$$

that is, $f(x)dx$ is the probability that X lies in the interval $x \leq X \leq x + dx$. For $f(x)$ to be a probability density, it has to satisfy $f(x) \geq 0$ and $\int f(x)dx = 1$.

- The generalization to probability distributions depending on several random variables is straightforward. Quantum physics abounds in examples.

Example 23.2.2 CONTINUOUS RANDOM VARIABLE: HYDROGEN ATOM

Quantum mechanics gives the probability $|\psi|^2 d^3r$ of finding a 1s electron in a hydrogen atom in volume³ d^3r , where $\psi = Ne^{-r/a}$ is the 1s wave function, a is the Bohr radius, and $N = (\pi a^3)^{-1/2}$ is a normalization constant such that

$$\int |\psi|^2 d^3r = 4\pi N^2 \int_0^\infty e^{-2r/a} r^2 dr = \pi a^3 N^2 = 1.$$

³Note that $|\psi|^2 4\pi r^2 dr$ gives the probability for the electron to be found between r and $r + dr$, at any angle.

The value of this integral can be checked by identifying it as a gamma function:

$$\int_0^{\infty} e^{-2r/a} r^2 dr = \left(\frac{a}{2}\right)^3 \int_0^{\infty} e^{-x} x^2 dx = \frac{a^3}{8} \Gamma(3) = \frac{a^3}{4}.$$

■

Computing Discrete Probability Distributions

In [Example 23.2.1](#) the overall probability of a particular value of a discrete random variable was computed as a product in which one factor was the number of equally likely ways in which that value could be obtained, and the other factor was the probability of each mutually exclusive occurrence. This type of computation arises sufficiently frequently that we should learn how to deal with it in general. Therefore, consider a situation in which N independent events take place (examples of such events include tosses of an individual die, selection of a card from a deck, energy state occupied by a molecule, orientation of the magnetic moment of a particle), and that each such event has one of a set of m mutually exclusive outcomes (e.g., number showing on the die, identity of the card, energy state, or magnetic moment orientation).

We assume that the outcomes $x_1, x_2, \dots, x_m$ of an individual event will have the respective probabilities $p_1, p_2, \dots, p_m$, with $p_1 + p_2 + \dots + p_m = 1$ (so that we have included all the possible outcomes). Then, we compute the probability that any n_1 of the events have outcome x_1 , any n_2 events have outcome x_2 , etc.:

$$P(n_1, n_2, \dots, n_m) = B(n_1, n_2, \dots, n_m) (p_1)^{n_1} (p_2)^{n_2} \dots (p_m)^{n_m}, \quad (23.17)$$

where $n_1 + n_2 + \dots + n_m = N$, and $B(n_1, n_2, \dots, n_m)$ is the number of ways that, for each i , n_i of the events have outcome x_i .

Now $B(n_1, n_2, \dots, n_m)$ is just the multinomial coefficient encountered earlier; in the present context the numbered objects correspond to events numbered from 1 to N and each box corresponds to an individual-event outcome. Thus, our final formula for the probability of a distribution defined by n_1, n_2 , etc., is

$$P(n_1, n_2, \dots, n_m) = \frac{N!}{n_1! n_2! \dots n_m!} (p_1)^{n_1} (p_2)^{n_2} \dots (p_m)^{n_m}. \quad (23.18)$$

Mean and Variance

When we make n measurements of a quantity x , obtaining the values x_j , we define the **average value**

$$\bar{x} = \frac{1}{n} \sum_{j=1}^n x_j \quad (23.19)$$

of the trials, also called the **mean** or **expectation value**, where this formula assumes that every observed value x_i is equally likely and occurs with probability $1/n$. This connection

is the key link of experimental data with probability theory. This observation and practical experience suggest defining the **mean value for a discrete random variable** X as

$$\langle X \rangle \equiv \sum_i x_i p_i, \quad (23.20)$$

while defining the mean value for a **continuous random variable** x characterized by probability density $f(x)$ as

$$\langle X \rangle = \int x f(x) dx. \quad (23.21)$$

Other notations for the mean in the literature are $\bar{X}$ and $E(X)$.

The use of the arithmetic mean $\bar{x}$ of n measurements as the average value is suggested by simplicity and plain experience, again assuming equal probability for each x_i . But why do we not consider the geometric mean

$$x_g = (x_1 x_2 \dots x_n)^{1/n}$$

or the harmonic mean x_h determined by the relation

$$\frac{1}{x_h} = \frac{1}{n} \left(\frac{1}{x_1} + \frac{1}{x_1} + \dots + \frac{1}{x_n} \right)$$

or the value $\tilde{x}$ that minimizes the sum of absolute deviations $|x_i - \tilde{x}|$? Here the x_i are taken to increase monotonically. When we plot $O(x) = \sum_{i=1}^{2n+1} |x_i - x|$, as in Fig. 23.4(a), for an odd number of points, we realize that it has a minimum at its central value $i = n$, while for an even number of points $E(x) = \sum_{i=1}^{2n} |x_i - x|$ is flat in its central region, as shown in Fig. 23.4(b). These properties make these functions unacceptable for determining average values. Instead, when we minimize (with respect to x) the sum of quadratic deviations,

$$\sum_{i=1}^n (x - x_i)^2 = \text{minimum}, \quad (23.22)$$

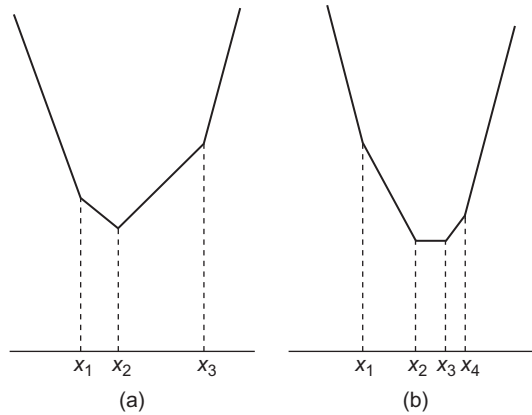


FIGURE 23.4 (a) $\sum_{i=1}^3 |x_i - x|$ for an odd number of points. (b) $\sum_{i=1}^4 |x_i - x|$ for an even number of points.

setting the derivative equal to zero yields $2 \sum_i (x - x_i) = 0$, or

$$x = \frac{1}{n} \sum_i x_i \equiv \bar{x},$$

that is, the arithmetic mean. The arithmetic mean has another important property: If we denote by $v_i = x_i - \bar{x}$ the deviations, then $\sum_i v_i = 0$, that is, the sum of positive deviations equals the sum of negative deviations. This principle of minimizing the quadratic sum of deviations, called the **method of least squares**, is due to C. F. Gauss, among others.

The ability of a mean value to represent a set of data points depends on the spread of the individual measurements from this mean. Again, we reject the average sum of deviations $\sum_{i=1}^n |x_i - \bar{x}|/n$ as a measure of the spread because it selects the central measurement as the best value for no good reason. A more appropriate definition of the spread is based on the average of the squares of the deviations from the mean. This quantity, known as the **standard deviation**, is defined as

$$\sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}, \quad (23.23)$$

where the square root is motivated by dimensional analysis.

If we square Eq. (23.23) and expand $(x_i - \bar{x})^2$, written as $(x_i - \langle x \rangle)^2$, we get

$$\begin{aligned} n\sigma^2 &= \sum_{i=1}^n x_i^2 - 2\langle x \rangle \sum_{i=1}^n x_i + n\langle x \rangle^2 \\ &= n(\langle x^2 \rangle - \langle x \rangle^2). \end{aligned}$$

Dividing through by n , we obtain the very useful formula,

$$\sigma^2 = \langle x^2 \rangle - \langle x \rangle^2. \quad (23.24)$$

Note that these two expectation values are equal only if all the x_i have the same value; for example, if we have two x_i , equal, respectively, to $\langle x \rangle + \delta$ and $\langle x \rangle - \delta$, then $\langle x^2 \rangle = \langle x \rangle^2 + \delta^2$, so the spread in the x_i has caused $\langle x_i^2 \rangle$ to increase.

Example 23.2.3 STANDARD DEVIATION OF MEASUREMENTS

From the measurements $x_1 = 7$, $x_2 = 9$, $x_3 = 10$, $x_4 = 11$, $x_5 = 13$, we extract $\bar{x} = 10$ for the mean value and, using Eq. (23.23),

$$\sigma = \sqrt{\frac{(-3)^2 + (-1)^2 + 0^2 + 1^2 + 3^2}{5}} = 2.2361$$

for the standard deviation, or spread. ■

There is yet another interpretation of the standard deviation, in terms of the sum of squares of measurement differences:

$$\begin{aligned}\sum_{i < k} (x_i - x_k)^2 &= \frac{1}{2} \sum_{i=1}^n \sum_{k=1}^n (x_i^2 + x_k^2 - 2x_i x_k) \\ &= \frac{1}{2} [2n^2 \langle x^2 \rangle - 2n^2 \langle x \rangle^2] = n^2 \sigma^2.\end{aligned}\quad (23.25)$$

The last step in the above equation made use of Eq. (23.24).

Now we are ready to generalize the spread in a set of n measurements with equal probability $1/n$ to the **variance** of an arbitrary probability distribution. For a discrete random variable X with probabilities p_i at $X = x_i$, we define the **variance**

$$\sigma^2 = \sum_j (x_j - \langle X \rangle)^2 p_j; \quad (23.26)$$

for a continuous probability distribution the definition becomes

$$\sigma^2 = \int_{-\infty}^{\infty} (x - \langle X \rangle)^2 f(x) dx. \quad (23.27)$$

We now develop some relationships satisfied by random variables:

1. The variance σ^2 of a random variable X has the property

$$\sigma^2 = \langle X^2 \rangle - \langle X \rangle^2. \quad (23.28)$$

This formula, previously derived as Eq. (23.24) only for a discrete random variable with all x_i equally probable, is true in general. The proof is left as Exercise 23.2.3.

2. If random variables X and Y are related by the linear equation $Y = aX + b$, then Y has mean value $\langle Y \rangle = a\langle X \rangle + b$ and variance $\sigma^2(Y) = a^2\sigma^2(X)$.

We prove this theorem only for a continuous distribution, leaving the case of a discrete random variable as an exercise for the reader. Directly from the definitions, we have

$$\langle Y \rangle = \int_{-\infty}^{\infty} (ax + b) f(x) dx = a\langle X \rangle + b,$$

where the integral multiplying b simplifies because $\int f(x) dx = 1$. For the variance we similarly obtain

$$\begin{aligned}\sigma^2(Y) &= \int_{-\infty}^{\infty} (ax + b - a\langle X \rangle - b)^2 f(x) dx = \int_{-\infty}^{\infty} a^2 (x - \langle X \rangle)^2 f(x) dx \\ &= a^2 \sigma^2(X).\end{aligned}$$

3. Probabilities of random variables satisfy the **Chebyshev inequality**,

$$P(|x - \langle X \rangle| \geq k\sigma) \leq \frac{1}{k^2}, \quad (23.29)$$

which demonstrates why the standard deviation serves as a measure of the spread of an arbitrary probability distribution from its mean value $\langle X \rangle$. We first derive the simpler inequality

$$P(Y \geq K) \leq \frac{\langle Y \rangle}{K}$$

for a continuous random variable Y with values y restricted to $y \geq 0$. (The proof for a discrete random variable follows along similar lines.) This inequality follows from

$$\begin{aligned} \langle Y \rangle &= \int_0^{\infty} yf(y)dy = \int_0^K yf(y)dy + \int_K^{\infty} yf(y)dy \\ &\geq \int_K^{\infty} yf(y)dy \geq K \int_K^{\infty} f(y)dy = KP(Y \geq K). \end{aligned}$$

Next we apply the same method to the positive variance integral,

$$\begin{aligned} \sigma^2 &= \int (x - \langle X \rangle)^2 f(x)dx \geq \int_{|x - \langle X \rangle| \geq k\sigma} (x - \langle X \rangle)^2 f(x)dx \\ &\geq k^2\sigma^2 \int_{|x - \langle X \rangle| \geq k\sigma} f(x)dx = k^2\sigma^2 P(|x - \langle X \rangle| \geq k\sigma), \end{aligned}$$

where we have first decreased the right-hand side by omitting the part of the positive integral with $|x - \langle X \rangle| \leq k\sigma$ and then decreased it further by replacing $(x - \langle X \rangle)^2$ in the remaining integral by its minimum value, $k^2\sigma^2$. We now divide the first and last members of this sequence of inequalities by the positive quantity $k^2\sigma^2$, thereby proving the Chebyshev inequality. For $k = 3$ we have the conventional three-standard-deviation estimate,

$$P(|x - \langle X \rangle| \geq 3\sigma) \leq \frac{1}{3^2} = \frac{1}{9}. \quad (23.30)$$

Moments of Probability Distributions

It is straightforward to generalize the mean value to higher moments of probability distributions relative to the mean value $\langle X \rangle$:

$$\begin{aligned} \langle (X - \langle X \rangle)^k \rangle &= \sum_j (x_j - \langle X \rangle)^k p_j, & \text{discrete distribution,} \\ \langle (X - \langle X \rangle)^k \rangle &= \int_{-\infty}^{\infty} (x - \langle X \rangle)^k f(x) dx, & \text{continuous distribution.} \end{aligned} \quad (23.31)$$

The **moment-generating function**

$$\langle e^{tX} \rangle = \int e^{tx} f(x) dx = 1 + t \langle X \rangle + \frac{t^2}{2!} \langle X^2 \rangle + \dots \quad (23.32)$$

is a weighted sum of the moments of the continuous random variable X , which is obtained by substituting the Taylor expansion of the exponential functions. Therefore,

$$\langle X \rangle = \left. \frac{d \langle e^{tX} \rangle}{dt} \right|_{t=0}, \quad \langle X^2 \rangle = \left. \frac{d^2 \langle e^{tX} \rangle}{dt^2} \right|_{t=0}, \dots, \quad \langle X^n \rangle = \left. \frac{d^n \langle e^{tX} \rangle}{dt^n} \right|_{t=0}. \quad (23.33)$$

Note that the moments here are not relative to the expectation value, but are relative to $x = 0$; they are called **central moments**.

Example 23.2.4 MOMENT-GENERATING FUNCTION

Suppose we have four cards, numbered from 1 through 4, from which we draw two at random and add their numbers. Letting this sum of the drawn numbers be values of a random variable X , we find that X has the following values and respective probabilities $P(x)$:

$$P(3) = 1/6, \quad P(4) = 1/6, \quad P(5) = 1/3, \quad P(6) = 1/6, \quad P(7) = 1/6.$$

Verifying these probabilities is the topic of [Exercise 23.2.1](#).

The moment-generating function for this system has the form

$$M = \frac{1}{6} (e^{3t} + e^{4t} + 2e^{5t} + e^{6t} + e^{7t}),$$

and its first two derivatives are

$$\begin{aligned} M' &= \frac{1}{6} (3e^{3t} + 4e^{4t} + 10e^{5t} + 6e^{6t} + 7e^{7t}), \\ M'' &= \frac{1}{6} (9e^{3t} + 16e^{4t} + 50e^{5t} + 36e^{6t} + 49e^{7t}). \end{aligned}$$

Setting $t = 0$, we get

$$\langle X \rangle = M'(0) = 5, \quad \langle X^2 \rangle = M''(0) = \frac{80}{3}.$$

Thus, the mean of X is found to be 5, and its variance is given by

$$\sigma^2 = \langle X^2 \rangle - \langle X \rangle^2 = \frac{80}{3} - 25 = \frac{5}{3}.$$

In this example we see that the moment-generating function does (in a systematic way) the same thing as direct formation of the moments; in a later example, [Example 23.3.2](#), we see a situation in which the use of the moment-generating function provides an opportunity to compute moments with rather little computational work. ■

Mean values, central moments, and variance can be defined analogously for probability distributions that depend on several random variables. We illustrate for the case of two random variables X and Y , for which the mean values and the variance of each variable take the forms

$$\langle X \rangle = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x, y) dx dy, \tag{23.34}$$

$$\langle Y \rangle = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(x, y) dx dy,$$

$$\sigma^2(X) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \langle X \rangle)^2 f(x, y) dx dy, \tag{23.35}$$

$$\sigma^2(Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (y - \langle Y \rangle)^2 f(x, y) dx dy.$$

Covariance and Correlation

Two random variables are said to be **independent** if the probability density $f(x, y)$ **factorizes** into a product $f(x)g(y)$ of probability distributions of one random variable each.

The covariance, defined as

$$\text{cov}(X, Y) = \langle (X - \langle X \rangle)(Y - \langle Y \rangle) \rangle, \tag{23.36}$$

is a measure of how much the random variables X and Y are correlated (or related): It is zero for independent random variables because

$$\begin{aligned}\text{cov}(X, Y) &= \int (x - \langle X \rangle) (y - \langle Y \rangle) f(x, y) dx dy \\ &= \int (x - \langle X \rangle) f(x) dx \int (y - \langle Y \rangle) g(y) dy \\ &= (\langle X \rangle - \langle X \rangle) (\langle Y \rangle - \langle Y \rangle) = 0.\end{aligned}$$

The normalized covariance $\text{cov}(X, Y)/\sigma(X)\sigma(Y)$, which has values between -1 and $+1$, is often called **correlation**.

In order to demonstrate that the correlation is bounded by

$$-1 \leq \frac{\text{cov}(X, Y)}{\sigma(X)\sigma(Y)} \leq 1,$$

we analyze the positive mean value

$$\begin{aligned}Q &= \langle [a(X - \langle X \rangle) + c(Y - \langle Y \rangle)]^2 \rangle \\ &= a^2 \langle [X - \langle X \rangle]^2 \rangle + 2ac \langle [X - \langle X \rangle][Y - \langle Y \rangle] \rangle + c^2 \langle [Y - \langle Y \rangle]^2 \rangle \\ &= a^2 \sigma(X)^2 + 2ac \text{cov}(X, Y) + c^2 \sigma(Y)^2 \geq 0.\end{aligned}\tag{23.37}$$

For this quadratic form to be nonnegative for all values of the constants a and c , its discriminant must satisfy $\text{cov}(X, Y)^2 - \sigma(X)^2 \sigma(Y)^2 \leq 0$, which proves the desired inequality.

The usefulness of the correlation as a quantitative measure is emphasized by the following theorem:

The probability $P(Y = aX + b)$ will be unity if, and only if, the correlation $\text{cov}(X, Y)/\sigma(X)\sigma(Y)$ is equal to ± 1 .

This theorem states that a $\pm 100\%$ correlation between X and Y implies not only some functional relation between both random variables but that the relation between them is linear.

Our first step in proving this theorem is to show that $P(Y = aX + b) = 1$ (meaning that $Y = aX + b$) implies that $\text{cov}(X, Y)/\sigma(X)\sigma(Y) = \pm 1$. For the mean $\langle Y \rangle$, we simply compute

$$\langle Y \rangle = \langle aX + b \rangle = a\langle X \rangle + b.$$

For the variance,

$$\begin{aligned}\sigma(Y)^2 &= \langle Y^2 \rangle - \langle Y \rangle^2 = \langle (aX + b)^2 \rangle - (a\langle X \rangle + b)^2 \\ &= a^2 \langle X^2 \rangle + 2ab\langle X \rangle + b^2 - (a^2 \langle X \rangle^2 + 2ab\langle X \rangle + b^2) \\ &= a^2 (\langle X^2 \rangle - \langle X \rangle^2) = a^2 \sigma(X)^2,\end{aligned}$$

which is equivalent to $\sigma(Y) = \pm a\sigma(X)$. We also need $\text{cov}(X, Y)$, which is

$$\begin{aligned}\text{cov}(X, Y) &= \langle (X - \langle X \rangle) ((aX + b) - (a\langle X \rangle + b)) \rangle \\ &= a \left(\langle X^2 \rangle - \langle X \rangle^2 \right) = a\sigma^2(X) = \pm \sigma(X)\sigma(Y),\end{aligned}$$

where the last equality was obtained by identifying $a\sigma(X)$ as $\pm\sigma(Y)$. This result completes the first step in our proof of the theorem.

To complete the proof, we must establish the converse of the relation we have just proved, namely that $\text{cov}(X, Y)/\sigma(X)\sigma(Y) = \pm 1$ implies $P(Y = aX + b) = 1$ for some set of values (a, b) . We proceed by forming the quadratic expectation value

$$\left\langle \left[(\sigma(Y)X \mp \sigma(X)Y) - \langle \sigma(Y)X \mp \sigma(X)Y \rangle \right]^2 \right\rangle,$$

where the symbol $\mp$ indicates that we choose a sign opposite to that of the correlation $\text{cov}(X, Y)/\sigma(X)\sigma(Y)$. Our plan is to show that this expectation value is zero. Since the expectation value is that of an inherently nonnegative quantity, we may then conclude that $\sigma(Y)X \mp \sigma(X)Y$ is (almost) everywhere equal to its expectation value, the value of which is some constant C . We therefore have

$$\sigma(Y)X \mp \sigma(X)Y = C, \quad \text{equivalent to} \quad Y = \pm \frac{\sigma(Y)X - C}{\sigma(X)},$$

the linear relation we seek.

It remains to confirm that the quadratic expectation value vanishes. Rearranging it first to the form

$$\left\langle \left[\sigma(Y)(X - \langle X \rangle) \mp \sigma(X)(Y - \langle Y \rangle) \right]^2 \right\rangle$$

and then expanding the square, we reach

$$\left\langle \sigma(Y)^2(X - \langle X \rangle)^2 + \sigma(X)^2(Y - \langle Y \rangle)^2 \mp 2\sigma(X)\sigma(Y)(X - \langle X \rangle)(Y - \langle Y \rangle) \right\rangle.$$

Making now the substitutions $(X - \langle X \rangle)^2 = \sigma(X)^2$, $(Y - \langle Y \rangle)^2 = \sigma(Y)^2$, and $\langle (X - \langle X \rangle)(Y - \langle Y \rangle) \rangle = \pm \sigma(X)\sigma(Y)$, our quadratic expectation value reduces to zero.

Marginal Probability Distributions

It is sometimes useful to integrate out (i.e., average over) one of the random variables in a multivariable distribution. When we do so, we are left with the probability distribution of the other random variables. For a two-variable distribution, we can eliminate either of the two variables:

$$F(x) = \int f(x, y)dy, \quad \text{or} \quad G(y) = \int f(x, y)dx, \quad (23.38)$$

and analogously for discrete probability distributions. When one or more random variables are integrated out, the remaining probability distribution is called **marginal**, the name

motivated by the geometric aspects of projection. It is straightforward to show that these marginal distributions satisfy all the requirements of properly normalized probability distributions.

Here is a comprehensive example that illustrates the computation of probability distributions and their mean values, variances, covariance, and correlation.

Example 23.2.5 REPEATED DRAWS OF CARDS

This example deals with independent repeated draws from a deck of playing cards. To make sure that these events stay independent, we draw the first card at random from a bridge deck containing 52 cards and then put it back at a random place and reshuffle the deck. Now we repeat the process for a second card. Let's define the random variables:

- X = number of so-called honors, that is, tens, jacks, queens, kings, or aces;
- Y = number of twos or threes.

In a single draw the probability of Event a (drawing an honor) is $p_a = 20/52 = 5/13$, while the probability of Event b (drawing a two or three) is $p_b = 2(4/52) = 2/13$. The probability of Event c (drawing anything else) is $p_c = (13 - 5 - 2)/13 = 6/13$. Since that exhausts all the mutually exclusive possibilities, we have $a + b + c = 1$.

In two drawings, it is possible to draw zero, one, or two honors (i.e., $x = 0, 1$, or 2). Likewise, we may draw zero, one, or two cards of value 2 or 3 (i.e., $y = 0, 1$, or 2). But because we are only drawing two cards, we have the additional condition $0 \leq x + y \leq 2$.

The probability function $P(X = x, Y = y)$, which we will write in the simpler form $P(x, y)$, is given by a formula of the type presented in Eq. (23.18), with N (the number of events) equal to 2 and with the three individual-event probabilities p_a , p_b , and p_c . The number of events a is x , the number of events b is y , and therefore the number of events c is $2 - x - y$, and, by Eq. (23.18),

$$\begin{aligned} P(x, y) &= \frac{2!}{x! y! (2 - x - y)!} (p_a)^x (p_b)^y (p_c)^{2-x-y} \\ &= \frac{2!}{x! y! (2 - x - y)!} \left(\frac{5}{13}\right)^x \left(\frac{2}{13}\right)^y \left(\frac{6}{13}\right)^{2-x-y}, \end{aligned} \quad (23.39)$$

with $0 \leq x + y \leq 2$. More explicitly, $P(x, y)$ has the following values:

$$\begin{aligned} P(0, 0) &= \left(\frac{6}{13}\right)^2, & P(1, 0) &= 2 \cdot \frac{5}{13} \cdot \frac{6}{13} = \frac{60}{13^2}, \\ P(2, 0) &= \left(\frac{5}{13}\right)^2, & P(0, 1) &= 2 \cdot \frac{2}{13} \cdot \frac{6}{13} = \frac{24}{13^2}, \\ P(0, 2) &= \left(\frac{2}{13}\right)^2, & P(1, 1) &= 2 \cdot \frac{5}{13} \cdot \frac{2}{13} = \frac{20}{13^2}. \end{aligned}$$

The probability distribution is properly normalized. Its expectation values are given by

$$\begin{aligned}\langle X \rangle &= \sum_{0 \leq x+y \leq 2} x P(x, y) = P(1, 0) + P(1, 1) + 2P(2, 0) \\ &= \frac{60}{13^2} + \frac{20}{13^2} + 2 \left(\frac{5}{13} \right)^2 = \frac{130}{13^2} = \frac{10}{13} = 2p_a,\end{aligned}$$

and

$$\begin{aligned}\langle Y \rangle &= \sum_{0 \leq x+y \leq 2} y P(x, y) = P(0, 1) + P(1, 1) + 2P(0, 2) \\ &= \frac{24}{13^2} + \frac{20}{13^2} + 2 \left(\frac{2}{13} \right)^2 = \frac{52}{13^2} = \frac{4}{13} = 2p_b.\end{aligned}$$

The values $2p_a$ and $2p_b$ are expected because we are drawing a card two times. The variances are

$$\begin{aligned}\sigma^2(X) &= \sum_{0 \leq x+y \leq 2} \left(x - \frac{10}{13} \right)^2 P(x, y) \\ &= \left(-\frac{10}{13} \right)^2 [P(0, 0) + P(0, 1) + P(0, 2)] + \left(\frac{3}{13} \right)^2 [P(1, 0) + P(1, 1)] + \left(\frac{16}{13} \right)^2 P(2, 0) \\ &= \frac{10^2 \cdot 64 + 3^2 \cdot 80 + 16^2 \cdot 5^2}{13^4} = \frac{4^2 \cdot 5 \cdot 169}{13^4} = \frac{80}{13^2},\end{aligned}$$

$$\begin{aligned}\sigma^2(Y) &= \sum_{0 \leq x+y \leq 2} \left(y - \frac{4}{13} \right)^2 P(x, y) \\ &= \left(-\frac{4}{13} \right)^2 [P(0, 0) + P(1, 0) + P(2, 0)] + \left(\frac{9}{13} \right)^2 [P(0, 1) + P(1, 1)] + \left(\frac{22}{13} \right)^2 P(0, 2) \\ &= \frac{4^2 \cdot 11^2 + 9^2 \cdot 44 + 22^2 \cdot 2^2}{13^4} = \frac{11 \cdot 4 \cdot 169}{13^4} = \frac{44}{13^2}.\end{aligned}$$

The covariance is given by

$$\begin{aligned}\text{cov}(X, Y) &= \sum_{0 \leq x+y \leq 2} \left(x - \frac{10}{13} \right) \left(y - \frac{4}{13} \right) P(x, y) = \frac{10 \cdot 4}{13^2} \cdot \frac{6^2}{13^2} - \frac{10 \cdot 9}{13^2} \cdot \frac{24}{13^2} \\ &\quad - \frac{10 \cdot 22}{13^2} \cdot \frac{4}{13^2} - \frac{3 \cdot 4}{13^2} \cdot \frac{60}{13^2} + \frac{3 \cdot 9}{13^2} \cdot \frac{20}{13^2} - \frac{16 \cdot 4}{13^2} \cdot \frac{5^2}{13^2} = -\frac{20}{13^2}.\end{aligned}$$

Therefore, the correlation of the random variables X, Y is given by

$$\frac{\text{cov}(X, Y)}{\sigma(X)\sigma(Y)} = -\frac{20}{8\sqrt{5} \cdot 11} = -\frac{1}{2}\sqrt{\frac{5}{11}} = -0.3371,$$

which means that there is a small (negative) correlation between these random variables, because if an honor is drawn, that drawing is not available to yield a 2 or a 3, and vice versa.

Finally, let us determine the marginal distribution,

$$P(X = x) = \sum_{y=0}^2 P(x, y),$$

or explicitly,

$$P(X = 0) = P(0, 0) + P(0, 1) + P(0, 2) = \left(\frac{6}{13}\right)^2 + \frac{24}{13^2} + \left(\frac{2}{13}\right)^2 = \left(\frac{8}{13}\right)^2,$$

$$P(X = 1) = P(1, 0) + P(1, 1) = \frac{60}{13^2} + \frac{20}{13^2} = \frac{80}{13^2},$$

$$P(X = 2) = P(2, 0) = \left(\frac{5}{13}\right)^2,$$

which is properly normalized because

$$P(X = 0) + P(X = 1) + P(X = 2) = \frac{64 + 80 + 25}{13^2} = \frac{169}{13^2} = 1.$$

The mean value and variance of X can be computed from the marginal probabilities:

$$\langle X \rangle = \sum_{x=0}^2 x P(X = x) = P(X = 1) + 2P(X = 2) = \frac{80 + 2 \cdot 25}{13^2} = \frac{130}{13^2} = \frac{10}{13},$$

$$\begin{aligned} \sigma_F^2 &= \sum_{x=0}^2 \left(x - \frac{10}{13}\right)^2 P(X = x) = \left(-\frac{10}{13}\right)^2 \left(\frac{8}{13}\right)^2 + \left(\frac{3}{13}\right)^2 \frac{80}{13^2} + \left(\frac{16}{13}\right)^2 \left(\frac{5}{13}\right)^2 \\ &= \frac{80 \cdot 169}{13^4} = \frac{80}{13^2}. \end{aligned}$$

These data agree with our earlier computations of the same quantities. ■

Conditional Probability Distributions

If we are interested in the distribution of a random variable X for a definite value $y = y_0$ of another random variable, then we deal with a **conditional probability distribution** $P(X = x | Y = y_0)$. The corresponding continuous probability density is $f(x, y_0)$.

Exercises

- 23.2.1** Verify the probabilities for the outcomes of the two-card draws in [Example 23.2.4](#), and by direct computation of the mean and variance check the results given in that example.

23.2.2 Show that adding a constant c to a random variable X changes the expectation value $\langle X \rangle$ by that same constant but not the variance. Show also that multiplying a random variable by a constant multiplies both the mean and variance by that constant. Show that the random variable $X - \langle X \rangle$ has mean value zero.

23.2.3 Using the definition given in Eq. (23.27) for the variance σ^2 of a continuous random variable, show that

$$\sigma^2 = \langle X^2 \rangle - \langle X \rangle^2.$$

23.2.4 A velocity $v_j = x_j/t_j$ is measured by recording the distances x_j at the corresponding times t_j . Show that $\bar{x}/\bar{t}$ is a good approximation for the average velocity v , provided that all the errors are small: $|x_j - \bar{x}| \ll |\bar{x}|$ and $|t_j - \bar{t}| \ll |\bar{t}|$.

23.2.5 Redefine the random variable Y in Example 23.2.5 as the number of fours through nines. Then determine the correlation of the X and Y random variables for the drawing of two cards (with replacement, as in the example).

23.2.6 The probability that a particle of an ideal gas travels a distance x between collisions is proportional to $e^{-x/f} dx$, where f is the constant mean free path. Verify that f is the average distance between collisions, and determine the probability of a free path of length $l \geq 3f$.

23.2.7 Determine the probability density for a particle in simple harmonic motion in the interval $-A \leq x \leq A$.

Hint. The probability that the particle is between x and $x + dx$ is proportional to the time it takes to travel across the interval.

23.3 BINOMIAL DISTRIBUTION

In this and the next two sections, we explore specific random variable distributions that are of importance both in physics and in the mathematical theories of probability and statistics. The topic of the present section is the **binomial distribution**, which typically occurs in the study of repeated independent trials of random events.

Example 23.3.1 REPEATED TOSSES OF DICE

What is the probability of three sixes in four tosses, all trials being independent? Getting one six in a single toss of a fair die has probability $a = 1/6$, and getting anything else has probability $b = 5/6$ with $a + b = 1$. Let the random variable $S = s$ be the number of sixes. In four tosses, $0 \leq s \leq 4$. The probability distribution $P(S = s)$ is given by the product of the two possibilities, a^s and b^{4-s} , times the number of ways that s sixes can be obtained from four tosses. This number is given by Eq. (23.18), and our probability is

$$P(S = s) = \frac{4!}{s!(4-s)!} a^s b^{4-s} = \binom{4}{s} a^s b^{4-s}. \quad (23.40)$$

We can now check that our probability is properly normalized by verifying that the sum of $P(S = s)$ for all s adds to unity. From properties of the binomial coefficients, we find

$$\sum_{s=0}^4 \binom{4}{s} a^s b^{4-s} = (a + b)^4 = \left(\frac{1}{6} + \frac{5}{6}\right)^4 = 1. \quad (23.41)$$

Writing out the cases of Eq. (23.40) explicitly, we have

$$f(0) = b^4, \quad f(1) = 4ab^3, \quad f(2) = 6a^2b^2, \quad f(3) = 4a^3b, \quad f(4) = a^4,$$

so we can answer our original question: The probability of three sixes in four tosses is

$$4a^3b = 4 \left(\frac{1}{6}\right)^3 \frac{5}{6} = \frac{5}{4 \cdot 3^4},$$

which is fairly small. ■

This case dealt with repeated independent trials, each with two possible outcomes of constant probability p for a hit and $q = 1 - p$ for a miss, and it is typical of many applications, including practical issues such as the random instances of defective products. The generalization to $S = s$ successes in n trials is given by the **binomial probability distribution**:

$$P(S = s) = \frac{n!}{s!(n-s)!} p^s q^{n-s} = \binom{n}{s} p^s q^{n-s}. \quad (23.42)$$

Figure 23.5 shows histograms for cases with 20 trials and various hit probabilities p .

Example 23.3.2 USE OF MOMENT-GENERATING FUNCTION

If we view our probability distribution as the result of adding together n random variables S_i , each having the value $s_i = 1$ with probability p and the value $s_i = 0$ with probability q , we can use the moment-generating function of Eq. (23.32) to obtain more information about the binomial distribution. We write

$$\langle e^{tS} \rangle = \langle e^{t(S_1 + S_2 + \dots + S_n)} \rangle = \langle e^{tS_1} \rangle \langle e^{tS_2} \rangle \dots \langle e^{tS_n} \rangle = \left[\langle e^{tS_1} \rangle \right]^n, \quad (23.43)$$

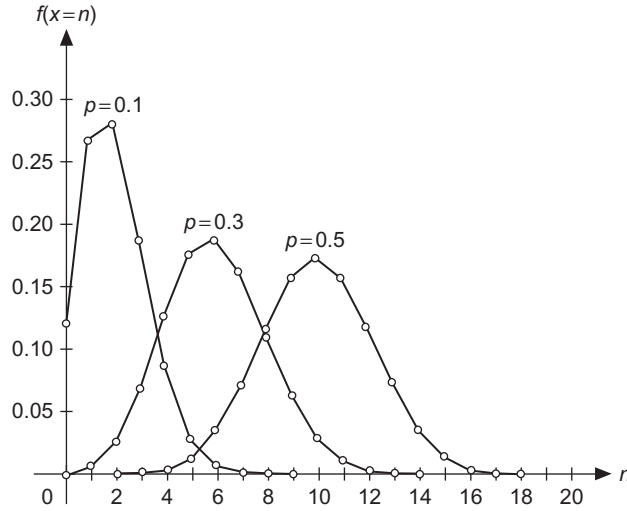
where we have used the fact that the trials are independent to write $\langle e^{tS} \rangle$ as a product of single-trial expectation values, all of which are identical.

We continue by evaluating $\langle e^{tS_1} \rangle$, which is an average for the two values $s_1 = 1$, with probability p , and $s_1 = 0$, with probability q . We get

$$\langle e^{tS_1} \rangle = pe^t + qe^0 = pe^t + q, \quad (23.44)$$

so Eq. (23.43) reduces to

$$\langle e^{tS} \rangle = (pe^t + q)^n. \quad (23.45)$$

FIGURE 23.5 Binomial probability distributions for $n = 20$ and $p = 0.1, 0.3, 0.5$.

Note that the fact the trials were independent enabled us to obtain the moment-distribution function without enumerating all the many-trial possibilities.

Now that we have $\langle e^{tS} \rangle$ we can differentiate it, as in Eq. (23.33), to obtain moments of our distribution. Using

$$\begin{aligned}\frac{\partial \langle e^{tS} \rangle}{\partial t} &= npe^t (pe^t + q)^{n-1}, \\ \langle S \rangle &= \sum_i s_i f(s_i) = \left. \frac{\partial \langle e^{tS} \rangle}{\partial t} \right|_{t=0} = np, \\ \frac{\partial^2 \langle e^{tS} \rangle}{\partial t^2} &= npe^t (pe^t + q)^{n-1} + n(n-1)p^2 e^{2t} (pe^t + q)^{n-2}, \\ \langle S^2 \rangle &= \sum_i s_i^2 f(s_i) = \left. \frac{\partial^2 \langle e^{tS} \rangle}{\partial t^2} \right|_{t=0} = np + n(n-1)p^2,\end{aligned}$$

we obtain, applying Eq. (23.28),

$$\begin{aligned}\sigma^2(S) &= \langle S^2 \rangle - \langle S \rangle^2 = np + n(n-1)p^2 - n^2 p^2 \\ &= np(1-p) = npq.\end{aligned}$$

For a given n , we see that the variance is largest when $p = q = 1/2$. This behavior is apparent in Fig. 23.5, where we see that the distribution broadens as p is increased from 0.1 to 0.3 to 0.5. ■

Exercises

- 23.3.1** Show that the variable $X = x$, defined as the number of heads in n coin tosses, is a random variable and determine its probability distribution. Describe the sample space. What are its mean value, the variance, and the standard deviation? Plot the probability function $P(x) = [n!/x!(n-x)!]2^{-n}$ for $n = 10, 20$, and 30 using graphics software.
- 23.3.2** Plot the binomial probability function for the probabilities $p = 1/6$, $q = 5/6$, and $n = 6$ throws of a die.
- 23.3.3** A hardware company knows that the probability of mass-producing nails includes a small probability $p = 0.03$ of defective nails (usually without a sharp tip). What is the probability of finding more than two defective nails in its commercial box of 100 nails?
- 23.3.4** Four cards are drawn from a shuffled bridge deck. What is the probability that they are all red? That they are all hearts? That they are honors? Compare the probabilities when each card is put back at a random place before drawing the next card, with the probabilities when the cards are not replaced in the deck.
- 23.3.5** Show that for the binomial distribution of Eq. (23.42), the most probable value of x is np .

23.4 POISSON DISTRIBUTION

The Poisson distribution is often used to describe situations in which an event occurs repeatedly at a constant rate of probability. Typical applications involve the decay of radioactive samples, but only in the approximation that the decay rate is slow enough that depletion in the population of the decaying species can be neglected. Other applications of interest include so-called **Poisson noise**, where fluctuations in a low rate of arrival of particles at a detector cause statistically predictable fluctuations in the detector signal.

The Poisson distribution can be developed by considering the probabilities that varying numbers of events are detected over an interval during which events occur at a constant rate of probability. The essential features of the development are that it assumes that (1) the event rate is small enough that there will be observationally accessible intervals in which at most one event occurs (i.e., one can consider intervals containing either zero or one event), and (2) the total number of events is small enough that it is useful to model their occurrence by a discrete probability distribution.

Let's proceed by defining the probability $P_n(t)$ that exactly n events occur in a time t , and that the probability of one event occurring in a short time interval dt will be μdt , where μ is a constant such that $\mu dt \ll 1$. This time interval dt is therefore short enough that we can neglect the possibility that more than one event occurs within it. Based on this hypothesis, we can set up a recursion relation for $P_n(t)$ by considering the two following mutually exclusive possibilities for the occurrence of n events in a time $d + dt$: (1) that n events occur during a time t and no events occur in a subsequent time interval dt , and (2) that $n - 1$ events occur during the time t and one event occurs during the subsequent interval dt . We therefore write

$$P_n(t + dt) = P_n(t)P_0(dt) + P_{n-1}(t)P_1(dt).$$

Then, inserting $P_1(dt) = \mu dt$ and $P_0(dt) = 1 - P_1(dt)$ and dividing through by dt , we get, after minor rearrangement,

$$\frac{dP_n(t)}{dt} = \frac{P_n(t+dt) - P_n(t)}{dt} = \mu P_{n-1}(t) - \mu P_n(t). \quad (23.46)$$

As a first step in solving this recursion relation, we note that for $n = 0$ it simplifies (because the possibility involving P_{n-1} does not exist) to

$$\frac{dP_0(t)}{dt} = -\mu P_0(t). \quad (23.47)$$

This equation, with initial condition $P_0(0) = 1$ (meaning that it is certain that no events are observed in an interval of zero length), has solution $P_0(t) = e^{-\mu t}$. Our solution informs us that the probability that no events have occurred before time t decays exponentially with t , at a rate dependent on the magnitude of μ . From this starting point and the further initial conditions $P_n(0) = 0$ for $n \geq 1$ (again, no detection of events occurs during an interval of zero length), the recursion relation can be solved to yield

$$P_n(t) = \frac{(\mu t)^n}{n!} e^{-\mu t}. \quad (23.48)$$

Equation (23.48) can be checked by substituting it into the recursion formula, Eq. (23.46), and by verifying that it satisfies the initial conditions $P_n(0) = \delta_{n0}$.

Equation (23.48) is taken as the definition of the Poisson distribution, regarded as a function of the quantity μt . Replacing μt by μ , we write the Poisson-distribution probabilities given for a discrete random variable X in the standard form,

$$p(n) = \frac{\mu^n}{n!} e^{-\mu}, \quad X = n = 0, 1, 2, \dots \quad (23.49)$$

We can check that the probabilities in Eq. (23.49) are properly normalized by noting that $\sum_n \mu^n/n!$ evaluates to e^μ . An example of a Poisson distribution is given in Fig. 23.6.

The mean value and variance of a Poisson distribution are easily calculated:

$$\langle X \rangle = \sum_{n=1}^{\infty} n \frac{\mu^n}{n!} e^{-\mu} = e^{-\mu} \sum_{n=1}^{\infty} \frac{\mu^n}{(n-1)!} = \mu, \quad (23.50)$$

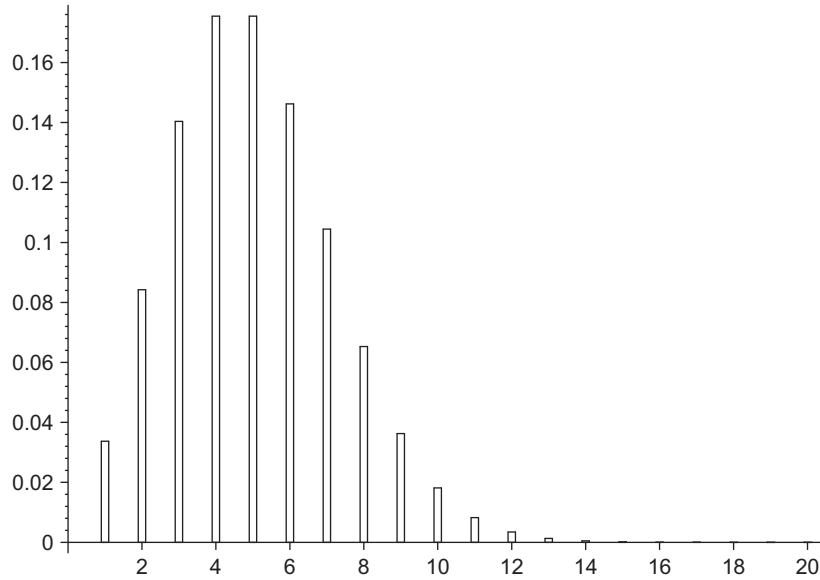
$$\langle X^2 \rangle = \sum_{n=1}^{\infty} n^2 \frac{\mu^n}{n!} e^{-\mu} = e^{-\mu} \sum_{n=1}^{\infty} \left[\frac{\mu^n}{(n-2)!} + \frac{\mu^n}{(n-1)!} \right] = \mu^2 + \mu, \quad (23.51)$$

$$\sigma^2 = \langle X^2 \rangle - \langle X \rangle^2 = \mu(\mu + 1) - \mu^2 = \mu. \quad (23.52)$$

The moments can also be calculated from the moment-generating function

$$\langle e^{tX} \rangle = \sum_{n=0}^{\infty} \frac{\mu^n}{n!} e^{-\mu} e^{tn} = e^{-\mu} \sum_{n=0}^{\infty} \frac{(\mu e^t)^n}{n!} = e^{\mu(e^t - 1)}.$$

Recall that the procedure for obtaining moments is to differentiate with respect to t and read out the derivatives evaluated at $t = 0$.

FIGURE 23.6 Poisson distribution, $\mu = 5$.

Relation to Binomial Distribution

A Poisson distribution becomes a good approximation of the binomial distribution for a large number n of trials and small probability $p \sim \mu/n$, with μ held constant.

Theorem: *In the limit $n \rightarrow \infty$ and $p \rightarrow 0$ so that the mean value $np \rightarrow \mu$ stays finite, the binomial distribution becomes a Poisson distribution.*

To prove this theorem, we need to find the large- n limit of the binomial distribution formula, Eq. (23.42). To do so, we apply Stirling's formula, in the form $n! \sim \sqrt{2\pi n}(n/e)^n$ for large n . See Eq. (12.110). For the quotient of the two n -dependent factorials occurring in Eq. (23.42), we have (keeping s finite while letting $n \rightarrow \infty$):

$$\frac{n!}{(n-s)!} \sim \left(\frac{n}{e}\right)^n \left(\frac{e}{n-s}\right)^{n-s} \sim \left(\frac{n}{e}\right)^s \left(\frac{n}{n-s}\right)^{n-s} \sim \left(\frac{n}{e}\right)^s \left(1 + \frac{s}{n-s}\right)^{n-s}.$$

The factor in the final expression raised to the power $n-s$ is, in the limit of large n , an expression of value e^s (in fact, it is, with $n-s$ changed to n , one of the often-used definitions of the exponential). The final result is

$$\frac{n!}{(n-s)!} \sim n^s. \quad (23.53)$$

We use a similar defining expression for the exponential to evaluate the factor q^{n-s} in Eq. (23.42). Writing $q^{n-s} = (1-p)^{n-s}$ and replacing p by its limiting value $p = \mu/n$,

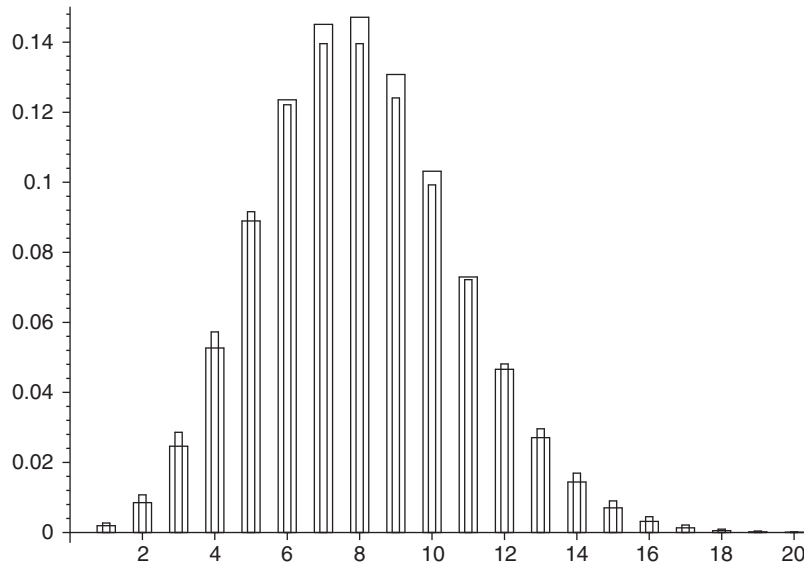


FIGURE 23.7 Comparison of binomial distribution ($N = 80$, $p = 0.1$), wide bars, and Poisson distribution ($\mu = 8$), narrow bars.

we have

$$q^{n-s} = (1-p)^{n-s} \sim \left(1 - \frac{\mu}{n}\right)^n \left(1 - \frac{\mu}{n}\right)^{-s} \sim e^{-\mu} (1) \sim e^{-\mu}. \quad (23.54)$$

Inserting the large- n limiting values from Eqs. (23.53) and (23.54) into the formula for the binomial distribution, we reach

$$P(S = s) = \frac{n!}{s!(n-s)!} p^s q^{n-s} \sim \frac{n!}{s!} p^s e^{-\mu} \sim \frac{\mu^s}{s!} e^{-\mu}, \quad (23.55)$$

where in the last step we have combined n^s and p^s into μ^s .

Equation (23.55) establishes our theorem, and thereby completes the connection between the Poisson and binomial distributions. This result, which becomes valid in the limit of a large number of trials, each of small probability, is sometimes referred to as an example of the **laws of large numbers**. A comparison of the binomial and Poisson distributions is presented as Fig. 23.7.

Exercises

- 23.4.1** Radioactive decays for long-lived isotopes are governed by the Poisson distribution. In a Rutherford-Geiger experiment, the numbers of emitted α particles are counted in each of $n = 2608$ time intervals of 7.5 seconds each. In Table 23.1 n_i is the number of time intervals in which i particles were emitted. Determine the average number λ of particles emitted per time interval, and compare the n_i of Table 23.1 with np_i computed from the Poisson distribution with mean value λ .

Table 23.1 Data for Exercise 23.4.1

i	$\rightarrow$	0	1	2	3	4	5	6	7	8	9	10
n_i	$\rightarrow$	57	203	383	525	532	408	273	139	45	27	16

- 23.4.2** Derive the standard deviation of a Poisson distribution of mean value μ .
- 23.4.3** The number of α -particles emitted by the decay of a radium sample is counted per minute for 40 hours. The total number is 5000. How many 1-minute intervals are expected with (a) 2, and (b) 5 α -particles?
- 23.4.4** For a radioactive sample, 10 decays are counted on average in 100 seconds. Use the Poisson distribution to estimate the probability of counting 3 decays in 10 seconds.
- 23.4.5** ^{238}U has a half-life of 4.51×10^9 years. Its decay series ends with the stable lead isotope ^{206}Pb . The ratio of the number of ^{206}Pb to ^{238}U atoms in a rock sample is measured as 0.0058. Estimate the age of the rock assuming that all the lead in the rock is from the initial decay of the ^{238}U , which determines the rate of the entire decay process, because the subsequent steps take place far more rapidly.
- Hint.* This is not a Poisson distribution problem, but is an application of the decay law $N(t) = N e^{-\lambda t}$, where λ , the decay constant, is related to the half-life T by $T = \ln 2 / \lambda$.
- ANS. 3.8×10^7 years.
- 23.4.6** The probability of hitting a target in one shot is known to be 20%. If five shots are fired independently, what is the probability of striking the target at least once?

23.5 GAUSS' NORMAL DISTRIBUTION

The bell-shaped Gauss distribution is defined by the probability density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{[x - \mu]^2}{2\sigma^2}\right), \quad -\infty < x < \infty, \quad (23.56)$$

with mean value μ and variance σ^2 . In part because it represents continuous limits of both the binomial and Poisson distributions, it is by far the most important continuous probability distribution and is displayed in Fig. 23.8.

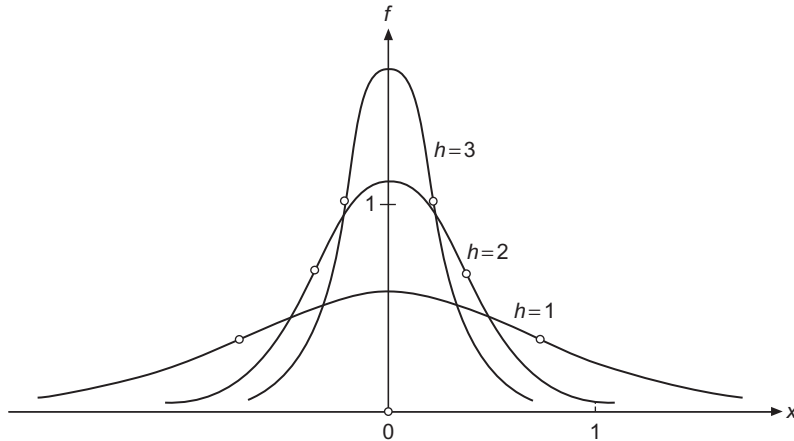


FIGURE 23.8 Gauss normal distribution for mean value zero and various standard deviations (marked by circles). Curves are labeled by $h = 1/\sigma\sqrt{2}$.

It is properly normalized because, substituting $y = (x - \mu)/\sigma\sqrt{2}$, we obtain

$$\frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x-\mu)^2/2\sigma^2} dx = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-y^2} dy = \frac{2}{\sqrt{\pi}} \int_0^{\infty} e^{-y^2} dy = 1.$$

To check the mean value, we can make the substitution $y = x - \mu$, and find that

$$\langle X \rangle - \mu = \int_{-\infty}^{\infty} \frac{x - \mu}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2} dx = \int_{-\infty}^{\infty} \frac{y}{\sigma\sqrt{2\pi}} e^{-y^2/2\sigma^2} dy = 0, \quad (23.57)$$

showing that $\langle X \rangle = \mu$. The zero result in Eq. (23.57) occurs because the integrand is odd in y , so the integral over $y > 0$ cancels that over $y < 0$. A check that the variance of this normal distribution is indeed σ^2 is the topic of Exercise 23.5.1.

We can compute conditional probabilities for the normal distribution. In particular, making for convenience the substitution $y = (x - \langle X \rangle)/\sigma$,

$$\begin{aligned} P(|X - \langle X \rangle| > k\sigma) &= P\left(\frac{|X - \langle X \rangle|}{\sigma} > k\right) = P(|Y| > k) \\ &= \sqrt{\frac{2}{\pi}} \int_k^{\infty} e^{-y^2/2} dy = \sqrt{\frac{4}{\pi}} \int_{k/\sqrt{2}}^{\infty} e^{-z^2} dz = \operatorname{erfc}(k/\sqrt{2}), \end{aligned}$$

we can evaluate the integral for $k = 1, 2, 3$, and thus extract the following numerical relations for a normally distributed random variable:

$$\begin{aligned} P(|X - \langle X \rangle| \geq \sigma) &\approx 0.3173, & P(|X - \langle X \rangle| \geq 2\sigma) &\approx 0.0455, \\ P(|X - \langle X \rangle| \geq 3\sigma) &\approx 0.0027. \end{aligned} \quad (23.58)$$

It is interesting to compare the last of these quantities with Chebyshev's inequality, which gives $1/9$ for the probability that an event falls further than 3σ from the mean. The $1/9$ applies to an **arbitrary** probability distribution, and is in strong contrast to the much smaller 0.0027 given by the 3σ -rule for the **normal** distribution.

Limits of Poisson and Binomial Distributions

In a special limit, the discrete Poisson probability distribution is closely related to the continuous Gauss distribution. This limit theorem is another example of the **laws of large numbers**, which are often dominated by the bell-shaped normal distribution.

Theorem: *For large n and mean value μ , the Poisson distribution approaches a Gauss distribution.*

To prove this theorem, in the limit $n \rightarrow \infty$, we approximate for large n the factorial in the Poisson probability $p(n)$ by Stirling's asymptotic formula, $n! \sim \sqrt{2n\pi} (n/e)^n$, and choose the deviation $v = n - \mu$ from the mean value as the new variable. We let the mean value μ approach ∞ and treat v/μ as small, but assume v^2/μ to be finite. Substituting $n = \mu + v$, we obtain

$$\begin{aligned} \ln p(n) &= \ln \left(\frac{\mu^n e^{-\mu}}{n!} \right) = n \ln \mu - \mu - \ln \sqrt{2n\pi} - n \ln n + n \\ &= (\mu + v) \ln \left(\frac{\mu}{\mu + v} \right) + v - \ln \sqrt{2(\mu + v)\pi} \\ &= (\mu + v) \ln \left(1 - \frac{v}{\mu + v} \right) + v - \ln \sqrt{2\pi(\mu + v)}. \end{aligned}$$

We next expand the first logarithmic term in powers of $v/(\mu + v)$, reaching

$$\ln p(n) = - \sum_{t=1}^{\infty} \frac{v^t}{t(\mu + v)^{t-1}} + v - \ln \sqrt{2\pi(\mu + v)}. \quad (23.59)$$

The first two terms of the t summation yield nonvanishing contributions in the large- μ limit; further terms vanish because the power of v in the numerator is less than twice that of μ in the denominator. Replacing $\mu + v$ by μ , [Eq. \(23.59\)](#) reduces to

$$\ln p(n) \sim -\frac{v^2}{2\mu} - \ln \sqrt{2\pi\mu}, \quad \text{equivalent to} \quad p(n) \sim \frac{1}{\sqrt{2\pi\mu}} e^{-v^2/2\mu}. \quad (23.60)$$

This is a Gauss distribution of the continuous variable v with mean value 0 and standard deviation $\sigma = \sqrt{\mu}$.

In another special limit, the discrete binomial probability distribution is also closely related to the continuous Gauss distribution. This limit theorem is yet another example of the **laws of large numbers**.

Theorem: In the limit $n \rightarrow \infty$, with p a finite trial probability such that the mean value $np \rightarrow \infty$, the binomial distribution becomes a Gauss normal distribution. Recall from [Section 23.4](#) that, when $np \rightarrow \mu < \infty$, the binomial distribution becomes a Poisson distribution.

Instead of the large number s of successes in n trials, we use the deviation $v = s - pn$ from the (large) mean value pn as our new continuous random variable, under the condition that as $n \rightarrow \infty$, $|v| \ll pn$ (so $v/n \rightarrow 0$) but v^2/n is finite. Thus, we replace s by $pn + v$ and $n - s$ by $qn - v$ in the factorials of the formula for the binomial distribution, [Eq. \(23.42\)](#). Writing now $W(v)$ as our probability distribution in the large- pn limit, we apply Stirling's formula as we have done several times before, obtaining initially

$$W(v) = \frac{p^s q^{n-s} n^{n+1/2} e^{-n+s+(n-s)}}{\sqrt{2\pi}(pn+v)^{s+1/2}(qn-v)^{n-s+1/2}}. \quad (23.61)$$

Next we factor out the dominant powers of n and cancel powers of p and q to find

$$W(v) = \frac{1}{\sqrt{2\pi pqn}} \left(1 + \frac{v}{pn}\right)^{-(pn+v+1/2)} \left(1 - \frac{v}{qn}\right)^{-(qn-v+1/2)}. \quad (23.62)$$

Taking the logarithm of $W(v)$ and expanding in powers of v , we retain only the terms through v^2 , yielding

$$\begin{aligned} \ln W(v) = & -\ln \sqrt{2\pi pqn} \\ & - \left[\frac{v}{n} \left(\frac{1}{2p} - \frac{1}{2q} \right) - \frac{v^2}{n^2} \left(\frac{1}{4p^2} + \frac{1}{4q^2} \right) + \frac{v^2}{n} \left(\frac{1}{2p} + \frac{1}{2q} \right) + \dots \right]. \end{aligned} \quad (23.63)$$

Setting v/n to zero, noting that

$$\left(\frac{1}{2p} + \frac{1}{2q} \right) = \frac{p+q}{2pq} = \frac{1}{2pq},$$

and dropping all terms v^t with $t > 2$, we obtain our large- n limit

$$W(v) = \frac{1}{\sqrt{2\pi pqn}} e^{-v^2/2pqn}, \quad (23.64)$$

which is a Gauss distribution in the deviations $s - pn$, with mean value 0 and standard deviation $\sigma = \sqrt{npq}$. The large values assumed for both pn and qn (and the discarded terms) restrict the validity of the theorem to the central part of the Gaussian bell shape, and exclude the tails.

Exercises

- 23.5.1** Show that the variance of the normal distribution given by [Eq. \(23.56\)](#) is σ^2 , the symbol in that equation.
- 23.5.2** Show that [Eq. \(23.62\)](#) can be obtained by manipulation of the formula [Eq. \(23.61\)](#) for $W(v)$.
- 23.5.3** With $W(v)$ the expression in [Eq. \(23.62\)](#), show that the expansion of $\ln W(v)$ in powers of v leads to [Eq. \(23.63\)](#).

- 23.5.4** What is the probability for a normally distributed random variable to differ by more than 4σ from its mean value? Compare your result with the corresponding one from Chebyshev's inequality. Explain the difference in your own words.
- 23.5.5** An instructor grades a final exam of a large undergraduate class, obtaining the mean value of points M and the variance σ^2 . Assuming a normal distribution for the number M of points, he defines a grade F when $M < m - 3\sigma/2$, D when $m - 3\sigma/2 < M < m - \sigma/2$, C when $m - \sigma/2 < M < m + \sigma/2$, B when $m + \sigma/2 < M < m + 3\sigma/2$, and A when $M > m + 3\sigma/2$. What is the percentage of As, Fs; Bs, Ds; and Cs? Redesign the cutoffs so that there are equal percentages of As and Fs (5%), 25% Bs and Ds, and 40% Cs.

23.6 TRANSFORMATIONS OF RANDOM VARIABLES

We have already encountered some elementary transformations involving random variables: In [Section 23.2](#) we observed that a random variable $Y = aX + b$ will have mean value $\langle Y \rangle = a\langle X \rangle + b$ and variance $\sigma^2(Y) = a^2\sigma^2(X)$. Here we consider more general transformations, with particular focus on continuous probability distributions.

First, consider a simple change of random variable from X to Y , where $y = y(x)$. If the probability distribution of X is $f(x)dx$, then the contribution at x to some quantity $M(y)$ is

$$P\{M[y(x)]\}dx = M[y(x)]f(x)dx. \quad (23.65)$$

But we may wish to express the probability in terms of the distribution of Y , writing

$$P[M(y)]dy = M(y)g(y)dy, \quad (23.66)$$

for y evaluated at the point corresponding to x , i.e., $y = y(x)$. To make these equations consistent, it is necessary that

$$g(y)dy = f(x)dx, \quad \text{or} \quad g(y) = f[x(y)]\frac{dx}{dy}. \quad (23.67)$$

For example, if $y = x^2$, then $dx/dy = 1/(2x) = y^{-1/2}/2$ and $g(y) = f(\sqrt{y})y^{-1/2}/2$.

Let's now address the transformation of two random variables X, Y into $U(X, Y), V(X, Y)$. Again we treat the continuous case. If

$$u = u(x, y), \quad v = v(x, y), \quad x = x(u, v), \quad y = y(u, v) \quad (23.68)$$

describe the transformation and its inverse; integrals of the probability density will transform by formulas that include the Jacobian of the transformation (see [Section 4.4](#)). The transformed probability density becomes

$$g(u, v) = f(x(u, v), y(u, v))|J|, \quad (23.69)$$

where the Jacobian is

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}. \quad (23.70)$$

This is a generalization of Eq. (23.67).

Addition of Random Variables

Let's apply this analysis to a situation in which Z is the sum of two random variables X and Y , or $Z = X + Y$. We transform to new variables X and Z , so the transformation is $x = x$, $z = x + y$; or $x = x$, $y = z - x$. The Jacobian for this transformation is

$$J = \begin{vmatrix} \frac{\partial x}{\partial x} & \frac{\partial x}{\partial z} \\ \frac{\partial(z-x)}{\partial x} & \frac{\partial(z-x)}{\partial z} \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ -1 & 1 \end{vmatrix} = 1.$$

If our original probability distribution was $f(x, y)$, it therefore transforms into $g(x, z) = f(x, z - x)$. We are usually interested in the marginal distribution in Z , obtained by integrating over x , and

$$P(Z = z) \equiv g(z) = \int_{-\infty}^{\infty} f(x, z - x) dx. \quad (23.71)$$

In the oft-occurring case that X and Y are independent random variables, so $f(x, y) = f_1(x)f_2(y)$, Eq. (23.71) assumes the form

$$g(z) = \int_{-\infty}^{\infty} f_1(x) f_2(z - x) dx, \quad (23.72)$$

which we recognize as a Fourier convolution, see Eq. (20.68):

$$g(z) = \int_{-\infty}^{\infty} f_1(x) f_2(z - x) dx = \sqrt{2\pi} (f_1 * f_2)(z). \quad (23.73)$$

Equation (23.72) gives us a general formula whereby we can obtain the distribution of $Z = X + Y$ from the distributions of independent variables X and Y , while Eq. (23.73) shows that it may be useful to consider the use of Fourier transforms for evaluating the integral. In fact, the moment-generating function, Eq. (23.32), is (if t is replaced by it) proportional to the Fourier transform of the probability density, and

$$\langle e^{itX} \rangle = \int e^{itx} f(x) dx = \sqrt{2\pi} f^T(t) \quad (23.74)$$

is known as the **characteristic function** in probability theory.

Applying the Fourier convolution theorem, Eq. (20.70), we therefore write

$$[P(Z = z)]^T(t) \equiv g^T(t) = \sqrt{2\pi} f_1^T(t) f_2^T(t), \quad (23.75)$$

showing that we can obtain $g(z)$ as the inverse Fourier transform

$$g(z) = \int e^{-izt} f_1^T(t) f_2^T(t) dt. \quad (23.76)$$

Connection with statistics texts will be improved by restating Eqs. (23.75) and (23.76) using the characteristic function notation. Equation (23.75) is equivalent to

$$\langle e^{itZ} \rangle = \langle e^{it(X+Y)} \rangle = \langle e^{itX} \rangle \langle e^{itY} \rangle. \quad (23.77)$$

Equation (23.76) states that $g(z)$ is the distribution that corresponds to $\langle e^{itZ} \rangle$; since Fourier transforms have inverses it can be assured that such a distribution exists.

Example 23.6.1 ADDITION THEOREM, NORMAL DISTRIBUTION

A good example of the analysis for a random variable $Z = X + Y$ is provided when X and Y are taken to be Gauss normal distributions with zero mean value and the same variance. This situation corresponds to a relationship known as the **addition theorem** for normal distributions.

Theorem: *If the independent random variables X, Y have identical normal distributions, that is, the same mean value and variance, then $Z = X + Y$ has normal distribution with twice the mean value and twice the variance of X and Y .*

To prove this theorem, we assume without loss of generality that the normal distributions each have variance $\sigma = 1$, so, from Eq. (23.56), each has the form

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-(x-\mu)^2/2}.$$

From Eq. (20.18) and the translation formula, Eq. (20.67), we find the Fourier transform of $f(x)$ to be

$$f^T(t) = \frac{1}{\sqrt{2\pi}} e^{it\mu} e^{-t^2/2}.$$

Now, applying Eq. (23.75), we have

$$g^T(t) = \frac{1}{\sqrt{2\pi}} e^{2it\mu} e^{-t^2}.$$

Taking the inverse transform, noting that the complex exponential shifts the origin by an amount 2μ , we get

$$g(z) = \frac{1}{2\sqrt{\pi}} e^{-(z-2\mu)^2/4},$$

which shows that the mean and variance of Z are twice those of X and Y , so the theorem is satisfied. ■

Multiplication or Division of Random Variables

Consider now the product $Z = XY$, taking X, Z as the new variables. This corresponds to the transformation $x = x, y = z/x$, with Jacobian

$$J = \begin{vmatrix} \frac{\partial x}{\partial x} & \frac{\partial x}{\partial z} \\ \frac{\partial(z/x)}{\partial x} & \frac{\partial(z/x)}{\partial z} \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ -z/x^2 & 1/x \end{vmatrix} = \frac{1}{x},$$

so the marginal distribution of Z is given by

$$g(z) = \int_{-\infty}^{\infty} f\left(x, \frac{z}{x}\right) \frac{dx}{|x|}. \quad (23.78)$$

If the random variables X, Y are independent with densities f_1, f_2 , then

$$g(z) = \int_{-\infty}^{\infty} f_1(x) f_2\left(\frac{z}{x}\right) \frac{dx}{|x|}. \quad (23.79)$$

Finally, let $Z = X/Y$, taking Y, Z as the new variables, corresponding to $x = yz, y = y$, with Jacobian

$$J = \begin{vmatrix} \frac{\partial(yz)}{\partial y} & \frac{\partial(yz)}{\partial z} \\ \frac{\partial y}{\partial y} & \frac{\partial y}{\partial z} \end{vmatrix} = \begin{vmatrix} z & y \\ 1 & 0 \end{vmatrix} = -y,$$

and the probability distribution of Z is given by

$$g(z) = \int_{-\infty}^{\infty} f(yz, y) |y| dy. \quad (23.80)$$

If the random variables X, Y are independent with densities f_1, f_2 , then

$$g(z) = \int_{-\infty}^{\infty} f_1(yz) f_2(y) |y| dy. \quad (23.81)$$

Gamma Distribution

Up to this point the only specific continuous probability distribution we have introduced is the Gauss normal distribution. However, if we make a change in the random variable of that distribution from X to $Y = X^2$, there will result a different distribution of significant utility, known as a **gamma distribution**. Let's start the present discussion with the now

quite familiar normal distribution of mean value zero and variance σ^2 . It has probability distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{x^2/2\sigma^2}.$$

As indicated just after Eq. (23.67), a transformation to write the distribution in terms of $y = x^2$ leads us to

$$g(y) = \frac{e^{-y/2\sigma^2}}{\sigma\sqrt{2\pi}} \frac{y^{-1/2}}{2}.$$

However, this equation does not take into account the fact that y must be restricted to nonnegative values, and that the same value of y will be encountered for two different values of x , namely $x = +\sqrt{y}$ and $x = -\sqrt{y}$. These considerations make a more proper and complete formula for $g(y)$ the following:

$$g(y) = \begin{cases} 0, & y \leq 0, \\ \frac{y^{-1/2} e^{-y/2\sigma^2}}{(2\sigma^2)^{1/2} \sqrt{\pi}}, & y > 0. \end{cases} \quad (23.82)$$

This expression for $g(y)$ is normalized (it must be, due to the way in which it was obtained). However, it is instructive to check, which is best done by changing to a new variable $z = y/2\sigma^2$, in terms of which we have

$$\int_0^\infty g(z) dz = \frac{1}{\sqrt{\pi}} \int_0^\infty z^{-1/2} e^{-z} dz = \frac{\Gamma(\frac{1}{2})}{\sqrt{\pi}} = 1,$$

where we have identified the integral as $\Gamma(\frac{1}{2})$ and also noted that $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.

Because the functional form of $g(y)$ is essentially that of the integrand of the integral representation of the gamma function, the distribution given by $g(y)$ is called a **gamma distribution**, and in particular, a gamma distribution with parameters $p = 1/2$ (the argument of the gamma function) and σ^2 (the variance of the underlying normal distribution). We generalize to gamma distributions of general p and σ :

$$g(p, \sigma; y) \equiv \begin{cases} 0, & y \leq 0, \\ \frac{y^{p-1} e^{-y/2\sigma^2}}{(2\sigma^2)^p \Gamma(p)}, & y > 0. \end{cases} \quad (23.83)$$

The gamma distribution often appears in contexts where the random variables involved need to be added together. It is therefore useful to take note of the Fourier transform of $g(p, \sigma; y)$:

$$[g(p, \sigma)]^T(t) = \frac{1}{\sqrt{2\pi}} \frac{1}{(1 - 2i\sigma^2 t)^p}. \quad (23.84)$$

Using the characteristic-function notation as introduced at Eq. (23.74), and defining X to be a (p, σ) gamma-distributed random variable, Eq. (23.84) takes the alternative form

$$\langle e^{itX} \rangle = \frac{1}{(1 - 2i\sigma^2 t)^p}. \quad (23.85)$$

Example 23.6.2 ADDITION OF GAMMA-DISTRIBUTION RANDOM VARIABLES

Let's compute the distribution of a random variable $Y = X_1 + X_2$, where X_1 has gamma distribution $g(p_1, \sigma; x_1)$ and X_2 has gamma distribution $g(p_2, \sigma; x_2)$. Note that both X_1 and X_2 have the same variance.

Using Eq. (23.77) for the characteristic function of $X_1 + X_2$ and Eq. (23.85) to evaluate $\langle e^{itX_j} \rangle$, we get

$$\langle e^{itY} \rangle = \frac{1}{(1 - 2i\sigma^2 t)^{p_1+p_2}}.$$

Recognizing this result as the characteristic function for a gamma distribution of parameter $p = p_1 + p_2$, we see that

$$g(y) = g(p_1 + p_2, \sigma; y).$$

Generalizing this result to an arbitrary number of X_j :

The probability distribution for a sum of gamma-distributed random variables X_j of parameters p_j but all of the same σ is a gamma distribution for that σ and with $p = \sum_j p_j$.

A corollary to the above is obtained if we consider the probability distribution of a sum of the form

$$Z = \sum_{j=1}^n X_j^2, \quad (23.86)$$

where the X_j are Gauss normal distributions, all with the same variance σ^2 . Because the quantities being summed are **squares** of random variables, it is useful first to make the substitutions $Y_j = X_j^2$, changing each distribution of X_j to that of a Y_j gamma distribution with $p = 1/2$, and finally combining the n gamma distributions to form the distribution of Z ; the result will be a gamma distribution with $p = n/2$ and the common value of σ . Summarizing,

The probability distribution for the sum of the squares of n Gauss normal random variables with a common variance σ^2 , as in Eq. (23.86), will be a gamma distribution with parameters $p = n/2$ and the common value of σ .

■

Exercises

- 23.6.1** Let $X_1, X_2, \dots, X_n$ be independent normal random variables with the same mean $\bar{x}$ and variance σ^2 . Show that

$$\frac{\sum_i X_i / n - \bar{x}}{\sigma \sqrt{n}}$$

is normal with mean zero and variance 1.

- 23.6.2** If the random variable X is normal with mean value 29 and standard deviation 3, what can you say about the distributions of $2X - 1$ and $3X + 2$?

- 23.6.3** For a normal distribution of mean value m and variance σ^2 , find the distance r such that half the area under the bell shape is between $m - r$ and $m + r$.
- 23.6.4** If $\langle X \rangle$, $\langle Y \rangle$ are the average values of two independent random variables X , Y , what is the expectation value of the product XY ?
- 23.6.5** If X and Y are two independent random variables with different probability densities and the function $f(x, y)$ has derivatives of any order, express $\langle f(X, Y) \rangle$ in terms of $\langle X \rangle$ and $\langle Y \rangle$. Develop similarly the covariance and correlation.
- 23.6.6** Let $f(x, y)$ be the joint probability density of two random variables X , Y . Find the variance $\sigma^2(aX + bY)$, where a, b are constants. What happens when X , Y are independent?
- 23.6.7** Obtain an addition theorem for the distribution of a random variable $Y = X_1 + X_2$ where X_1 and X_2 are Gauss normal distributions with different mean values μ_j and variances σ_j^2 .

ANS. Y is normal with mean $\mu_1 + \mu_2$ and variance $\sigma_1^2 + \sigma_2^2$.

- 23.6.8** Show that the Fourier transform of the gamma-distribution probability density, Eq. (23.83), has the functional form given in Eq. (23.84).

23.7 STATISTICS

In statistics, probability theory is applied to the evaluation of data from random experiments or to samples to test some hypothesis because the data have random fluctuations due to lack of complete control over the experimental conditions. Typically one attempts to estimate the mean value and variance of the distributions from which the samples derive, and to generalize properties valid for a sample to the rest of the events at a prescribed confidence level. Any assumption about an unknown probability distribution is called a **statistical hypothesis**. The concepts of tests and confidence intervals are among the most important developments of statistics.

Error Propagation

When we measure a quantity x repeatedly, obtaining the values x_j , or select a sample for testing, we can compute

$$\bar{x} = \frac{1}{n} \sum_{j=1}^n x_j, \quad \sigma^2 = \frac{1}{n} \sum_{j=1}^n (x_j - \bar{x})^2,$$

where $\bar{x}$ is the mean value and σ^2 is the variance, a measure of the spread of the points about the mean value. We can write $x_j = \bar{x} + e_j$, where e_j is the deviation from the mean value, and we know that $\sum_j e_j = 0$.

Now suppose we want to estimate the value of a known function $f(x)$ based on these measurements x_j ; that is, we want to assign a value of f given the set $f_j = f(x_j)$.

Substituting $x_j = \bar{x} + e_j$ and forming the mean value

$$\begin{aligned}\bar{f} &= \frac{1}{n} \sum_j f(x_j) = \frac{1}{n} \sum_j f(\bar{x} + e_j) \\ &= f(\bar{x}) + \frac{1}{n} f'(\bar{x}) \sum_j e_j + \frac{1}{2n} f''(\bar{x}) \sum_j e_j^2 + \cdots \\ &= f(\bar{x}) + \frac{1}{2} \sigma^2 f''(\bar{x}) + \cdots,\end{aligned}\quad (23.87)$$

we obtain the average value $\bar{f}$ as $f(\bar{x})$ in lowest order, as expected. But in second order there is a correction given by the variance with a scale factor $f''(\bar{x})/2$.

It is also of interest to determine the spread predicted for the values of $f(x_j)$. To lowest order, this is given by the average of the sum of squares of the deviations. Approximating f_j as $\bar{f} + f'(\bar{x})e_j$, we get

$$\sigma^2(f) \equiv \frac{1}{n} \sum_j (f_j - \bar{f})^2 \approx [f'(\bar{x})]^2 \frac{1}{n} \sum_j e_j^2 = [f'(\bar{x})]^2 \sigma^2. \quad (23.88)$$

In summary, we may formulate somewhat symbolically

$$f(\bar{x} \pm \sigma) = f(\bar{x}) \pm f'(\bar{x})\sigma$$

as the simplest form of error propagation for a function of one measured variable.

For a function $f(x_j, y_k) \equiv f_{jk}$ of two quantities $x_j = \bar{x} + u_j$, $y_k = \bar{y} + v_k$, where the x_j and y_k are measured independently of each other and we have r values of j and s values of k , we obtain similarly

$$\begin{aligned}\bar{f} &= \frac{1}{rs} \sum_{j=1}^r \sum_{k=1}^s f_{jk} = f(\bar{x}, \bar{y}) + \frac{1}{r} f_x \sum_j u_j + \frac{1}{s} f_y \sum_k v_k + \cdots \\ &\approx f(\bar{x}, \bar{y}).\end{aligned}\quad (23.89)$$

The error in $\bar{f}$ is seen to be second-order in the u_j and v_k . In writing [Eq. \(23.89\)](#) we have used the relations $\sum_j u_j = \sum_k v_k = 0$ and have introduced the definitions

$$f_x = \left. \frac{\partial f}{\partial x} \right|_{\bar{x}, \bar{y}}, \quad f_y = \left. \frac{\partial f}{\partial y} \right|_{\bar{x}, \bar{y}}. \quad (23.90)$$

The variance of f is (to first order)

$$\sigma^2(f) = \frac{1}{rs} \sum_{j=1}^r \sum_{k=1}^s (f_{jk} - \bar{f})^2 = \frac{1}{rs} \sum_{j,k} (u_j f_x + v_k f_y)^2 = \frac{f_x^2}{r} \sum_j u_j^2 + \frac{f_y^2}{s} \sum_k v_k^2,$$

where we have dropped the zero cross term $\sum_{j,k} u_j v_k = \sum_j u_j \sum_k v_k$. Noting that $\sum_j u_j^2 = r\sigma_x^2$ and $\sum_k v_k^2 = s\sigma_y^2$, we reach the final result

$$\sigma^2(f) = \frac{1}{rs} \sum_{j,k} (f_{jk} - \bar{f})^2 = f_x^2 \sigma_x^2 + f_y^2 \sigma_y^2. \quad (23.91)$$

Symbolically, the error propagation for a function of two measured variables may be summarized as

$$f(\bar{x} \pm \sigma_x, \bar{y} \pm \sigma_y) = f(\bar{x}, \bar{y}) \pm \sqrt{f_x^2 \sigma_x^2 + f_y^2 \sigma_y^2}.$$

Example 23.7.1 REPEATED MEASUREMENTS

As an application and generalization of the result given in Eq. (23.91), let's consider what happens when we regard the mean of n measurements x_j as a function

$$\bar{x} = f(x_1, x_2, \dots, x_n) = (x_1 + x_2 + \dots + x_n)/n$$

of the variables $x_1, \dots, x_n$, each with variance σ^2 . Then we have $f_{x_j} = 1/n$ for each j , and, according to Eq. (23.91),

$$\sigma^2(\bar{x}) = \sum_{j=1}^n f_{x_j}^2 \sigma^2 = \sum_{j=1}^n \frac{\sigma^2}{n^2} = \frac{\sigma^2}{n}. \quad (23.92)$$

This result indicates that the standard deviation of the mean value, $\sigma(\bar{x})$, will decrease with the number of repeated measurements, approaching zero as $\sigma/\sqrt{n}$. It is important to recognize the distinction between the variance of the mean value, denoted $\sigma^2(\bar{x})$, and the corresponding quantity for the individual measurements (denoted σ^2).

If we refer now to our earlier result that the sum of n identically distributed, Gauss normal random variables is also a normal random variable with a variance equal to n times that of each variable (see Example 23.6.1), and note also that division of the sum by n (to form the mean) causes division of the variance by n^2 , as discussed following Eq. (23.28), we find a result identical to that developed in the present example, but with the additional feature that the mean is also normally distributed. ■

The arithmetic mean $\bar{x}$ will, because of the distribution in the x_j , differ from the true (but unknown) value μ , with μ and $\bar{x}$ differing by some amount α , or $\bar{x} = \mu + \alpha$. However, as the number n of measurements increases, we expect that the error α will tend to zero, and that, according to Example 23.7.1, we can estimate α to fall in the range $(-\sigma/\sqrt{n} < \alpha < \sigma/\sqrt{n})$. We can refine this estimate by considering the spread of the x_j measured with respect to the true value μ , meaning that we compute the variance using the average of v_j^2 , where $v_j = x_j - \mu$, instead of that of e_j^2 , where, as before, $e_j = x_j - \bar{x}$. Calling this version of the variance s^2 , we write

$$s^2 = \frac{1}{n} \sum_{j=1}^n v_j^2 = \frac{1}{n} \sum_{j=1}^n (e_j + \alpha)^2 = \frac{1}{n} \sum_{j=1}^n e_j^2 + \alpha^2, \quad (23.93)$$

where the term linear in e_j vanishes because $\sum_j e_j = 0$. Inserting now an estimate of α , in the form $\alpha^2 \approx s^2/n$ (this approximation good to first order), Eq. (23.93) rearranges to

$$s^2 \left(1 - \frac{1}{n}\right) = \frac{1}{n} \sum_{j=1}^n e_j^2, \quad (23.94)$$

equivalent to

$$s = \sqrt{\frac{\sum_j (x_j - \bar{x})^2}{n-1}}. \quad (23.95)$$

The quantity s is referred to as the **sample standard deviation**. Equation (23.95) is not well defined when $n = 1$, but that is not an issue because a single data point is insufficient to determine a spread. The presence of $n - 1$, in contrast to the factor n in Eq. (23.23), allows for the probable error in $\bar{x}$, and is known as **Bessel's correction** to the standard deviation formula.

Fitting Curves to Data

Suppose we have a sample of measurements y_j taken at times t_j , where the time is known precisely but the y_j are subject to experimental error. An example would be snapshots of the position of a particle in uniform motion at the times t_j . Our statistical hypothesis, motivated by Newton's first law and the initial condition that $y = 0$ when $t = 0$, is that $y(t)$ satisfies an equation of the form $y = at$, where the constant a is to be determined from the measurements.

To fit our equation to the data, we first minimize the sum of the squares of deviations $S = \sum_j (at_j - y_j)^2$ to determine the slope parameter a , also called **regression coefficient**, using the method of least squares. Differentiating S with respect to a we obtain

$$2 \sum_j (at_j - y_j)t_j = 0,$$

which we can solve for a :

$$a = \frac{\sum_j t_j y_j}{\sum_j t_j^2}. \quad (23.96)$$

Note that the numerator is built like a sample covariance, the scalar product of the variables t , y of the sample. As shown in Fig. 23.9, the measured values y_j do not as a rule lie on the line. They have the **sample standard deviation**, computed from Eq. (23.95),

$$s = \sqrt{\frac{\sum_j (y_j - at_j)^2}{n-1}}.$$

Alternatively, suppose that the y_j values are known precisely while the t_j are measurements subject to experimental error. As suggested by Fig. 23.10, in this case we need to

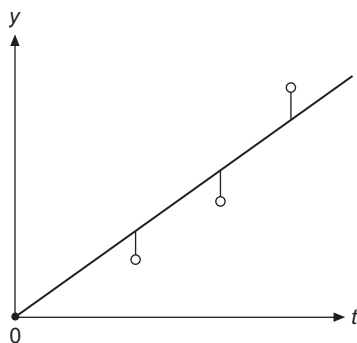


FIGURE 23.9 Straight line fit to data points (t_j, y_j) with t_j known, y_j measured.

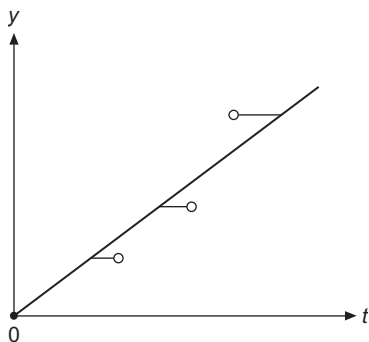


FIGURE 23.10 Straight line fit to data points (t_j, y_j) with y_j known, t_j measured.

interchange the roles of t and y and to fit the line $t = by$ to the data points. We minimize $S = \sum_j (by_j - t_j)^2$, setting $dS/db = 0$, and find similarly the slope parameter

$$b = \frac{\sum_j t_j y_j}{\sum_j y_j^2}. \quad (23.97)$$

In case both t_j and y_j have errors (we take t and y to have the same measurement precision), we have to minimize the sum of squares of the deviations of both variables. It is convenient to fit to a parameterization $t \sin \alpha - y \cos \alpha = 0$, so $y/t = \sin \alpha / \cos \alpha = \tan \alpha$, meaning that α is the angle the fitting line makes with the t -axis (see Fig. 23.11). Our task will therefore be to determine α . We also see from Fig. 23.11 that the fitting line has to be drawn so that the sum of the squares of the distances d_j of the points (t_j, y_j) from the line becomes a minimum. To find d_j , we rotate our coordinate system the angle α , which moves (t_j, y_j) to (t'_j, y'_j) according to

$$\begin{pmatrix} t'_j \\ y'_j \end{pmatrix} = \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix} \begin{pmatrix} t_j \\ y_j \end{pmatrix},$$

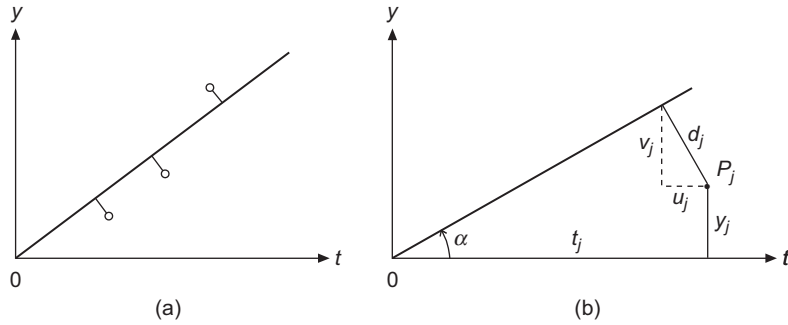


FIGURE 23.11 (a) Straight line fit to data points (t_j, y_j) . (b) Geometry of deviations u_j, v_j, d_j .

which yields $d_j = y'_j = -t_j \sin \alpha + y_j \cos \alpha$, the (signed) distance to the line at angle α . The minimum of the square of the distances to the line is found from

$$\begin{aligned} \frac{d}{d\alpha} \sum_j d_j^2 &= 2 \sum_j (-t_j \sin \alpha + y_j \cos \alpha)(-t_j \cos \alpha - y_j \sin \alpha) \\ &= \sin \alpha \cos \alpha \sum_j (t_j^2 - y_j^2) - (\cos^2 \alpha - \sin^2 \alpha) \sum_j t_j y_j = 0, \end{aligned}$$

which can be reduced to

$$\tan 2\alpha = \frac{2 \sum_j t_j y_j}{\sum_j (t_j^2 - y_j^2)}. \quad (23.98)$$

This least-squares fitting is appropriate when the measurement errors are unknown, as it gives equal weight to the deviation of each point from the fitting line.

Finally, if we have information that permits the assignment of different probable errors to different points, we have the alternative of making a “weighted” least-squares fit called a **chi square** fit, which we discuss in the next subsection.

The χ^2 Distribution

Given a set of u_j corresponding to values t_j of an independent variable (which is not necessarily a time), we seek to fit these data to a function $u(t, a_1, a_2, \dots)$, where the a_i are parameters that are adjusted to optimize the fit. The optimization is carried out by minimizing a **weighted sum** of the squares of the deviations, where the weights are controlled by the assumed standard deviations σ_j of the respective measurements u_j . The quantity to be minimized is traditionally labeled χ^2 and called **chi-square**, and its precise definition is

$$\chi^2 = \sum_{j=1}^n \left(\frac{u_j - u(t_j, a, \dots)}{\sigma_j} \right)^2, \quad (23.99)$$

where n is the number of data points. This quadratic merit function gives more weight to points with small measurement uncertainties σ_j .

The key assumptions adopted to analyze the probability distribution corresponding to the **chi-square** fit are (1) that each data point is an independent Gauss normal random variable X_j with zero mean and unit variance, with the unit variance assured by the presence of the σ_j in each term, and (2) that the distribution χ^2 is related to the X_j by

$$\chi^2 = \sum_{j=1}^n X_j^2. \quad (23.100)$$

Making a chi-square fit requires no knowledge of statistics; we simply apply standard analytical or numerical methods to minimize χ^2 for our set of data points. On the other hand, a knowledge of the chi-square probability distribution will be needed to determine whether we are getting the expected quality from our chi-square fit. In particular, if we wish to determine the probability of the occurrence of our data set based on the chi-square distribution (and possibly assess the adequacy of our assumptions regarding the individual-point variances σ_j^2), we must undertake further analysis.

Our earlier discussion of transformations of random variables included the analysis of sums of normally distributed X_j^2 of the form given in Eq. (23.100), with the result developed in Example 23.6.2. Specializing to the case at hand, we note that χ^2 will have a gamma probability distribution with parameters $p = n/2$ and $\sigma = 1$, so

$$g(\chi^2 = y) = \frac{y^{(n/2)-1} e^{-y/2}}{2^{n/2} \Gamma(n/2)}. \quad (23.101)$$

Plots of $g(y)$ for several values of n are given in Fig. 23.12.

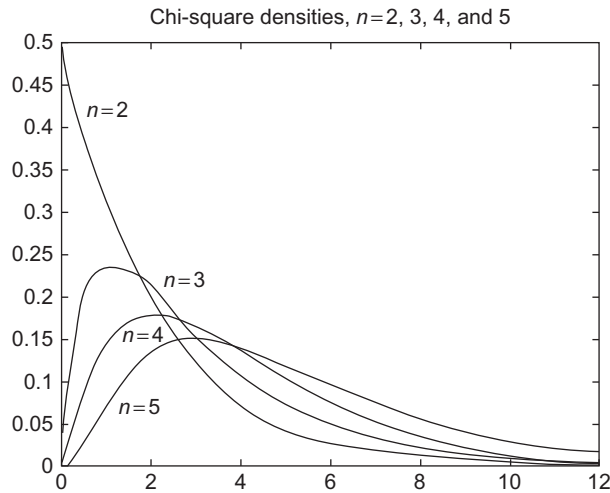


FIGURE 23.12 χ^2 probability density $g_n(y)$.

Table 23.2 χ^2 Distribution

n	$v = 0.8$	$v = 0.7$	$v = 0.5$	$v = 0.4$	$v = 0.3$	$v = 0.2$	$v = 0.1$
1	0.064	0.148	0.455	0.708	1.074	1.642	2.706
2	0.446	0.713	1.386	1.833	2.408	3.219	4.605
3	1.005	1.424	2.366	2.946	3.665	4.642	6.251
4	1.649	2.195	3.357	4.045	4.878	5.989	7.779
5	2.343	3.000	4.351	5.132	6.064	7.289	9.236
6	3.070	3.828	5.348	6.211	7.231	8.558	10.645

Note: A data set with n degrees of freedom will have probability v that its value of χ^2 exceeds the tabulated value.

It is also useful to note the moment-generating function for this distribution:

$$\langle e^{t\chi^2} \rangle = \frac{1}{(1 - 2t)^{n/2}}, \quad (23.102)$$

a result that follows directly from Eq. (23.85). Differentiating Eq. (23.102), we find

$$\langle \chi^2 \rangle = \frac{d(1 - 2t)^{-n/2}}{dt} \Big|_{t=0} = n, \quad \langle (\chi^2)^2 \rangle = \frac{d^2(1 - 2t)^{-n/2}}{dt^2} \Big|_{t=0} = n(n + 2), \quad (23.103)$$

and therefore

$$\sigma^2(\chi^2) = \langle (\chi^2)^2 \rangle - \langle \chi^2 \rangle^2 = n(n + 2) - n^2 = 2n. \quad (23.104)$$

These results suggest that typical data with realistically assigned individual-measurement variances would yield a value of χ^2 comparable to the number of data points. However, by calculating

$$P(\chi^2 > y_0) = \int_{y_0}^{\infty} g(y) dy, \quad (23.105)$$

where $g(y)$ is the distribution in Eq. (23.101), we can obtain for any y_0 the probability that a data set would have a larger spread than that corresponding to $\chi^2 = y_0$. Because it is somewhat laborious to compute the integral in Eq. (23.105), its values are generally obtained by table lookup. A short table of these chi-square data are given in Table 23.2.

Before closing this subsection, we need to deal with the fact that our random variables X_j do not really have zero mean values if the function $u(t_j, \dots)$ was chosen based on the available data and therefore was not exact. By reasoning similar to that involved in the discussion leading to Eq. (23.95), it can be shown that if a chi-square fit involves n data points and the determination of r parameters, the effective number of degrees of freedom is $n - r$, with the implication that the inexactness of the fit in r degrees of freedom corresponds to a chi-square distribution with n replaced by $n - r$.

Finally, it is worth pointing out that the χ^2 analysis does not really test the assumptions that the data points are independent normal random variables. If these assumptions are not approximately valid, it is unlikely that good chi-square fits can be achieved.

Example 23.7.2 CHI-SQUARE FIT

Let us apply the χ^2 function to a straight-line fit of the type shown in Fig. 23.9, with the three measured points and their individual standard deviations, written as $(t_j, u_j \pm \sigma_j)$, having the values

$$(1, 0.8 \pm 0.1), \quad (2, 1.5 \pm 0.05), \quad (3, 2.7 \pm 0.2).$$

Before proceeding to the chi-square fit, we first fit a line assuming the points to be equally weighted, corresponding to using Eq. (23.96) for the slope a . We find

$$a = \frac{1(0.8) + 2(1.5) + 3(2.7)}{1^2 + 2^2 + 3^2} = \frac{11.9}{14} = 0.850.$$

The sample variance of the points from the line is

$$\sigma^2 = \frac{1}{2} \left([0.8 - 1(0.850)]^2 + [1.5 - 2(0.850)]^2 + [2.7 - 3(0.850)]^2 \right) = 0.0325,$$

and the variance of a is

$$\sigma^2(a) = \sum_j \left(\frac{\partial a}{\partial u_j} \right)^2 \sigma_j^2 = \sum_j \left(\frac{t_j}{\sum_k t_k^2} \right)^2 \sigma_j^2 = \frac{\sigma^2}{\sum_k t_k^2} = \frac{0.0325}{14} = 0.00232.$$

Thus, the unweighted fit yields $a = 0.850 \pm \sqrt{0.00232} = 0.850 \pm 0.048$.

Turning now to the chi-square fit, we next minimize

$$\chi^2 = \sum_j \left(\frac{u_j - at_j}{\sigma_j} \right)^2$$

with respect to a . This process yields

$$\frac{\partial \chi^2}{\partial a} = -2 \sum_j \frac{t_j(u_j - at_j)}{\sigma_j^2} = 0,$$

or

$$a = \sum_j \frac{t_j u_j}{\sigma_j^2} \bigg/ \sum_j \frac{t_j^2}{\sigma_j^2}.$$

In our case

$$a = \frac{\frac{1(0.8)}{0.1^2} + \frac{2(1.5)}{0.05^2} + \frac{3(2.7)}{0.2^2}}{\frac{1^2}{0.1^2} + \frac{2^2}{0.05^2} + \frac{3^2}{0.2^2}} = \frac{1482.5}{1925} = 0.770.$$

The value we obtained for a is dominated by the middle point, the smallest σ_j ; if that point were the only one used, we would have gotten $a = 1.5/2 = 0.75$. The variance of a , $\sigma^2(a)$, is now

$$\sigma^2(a) = \sum_j \left(\frac{\partial a}{\partial u_j} \right)^2 \sigma_j^2 = \sum_j \left(\frac{t_j/\sigma_j^2}{\sum_k t_k^2/\sigma_k^2} \right)^2 \sigma_j^2 = \frac{1}{\sum_k t_k^2/\sigma_k^2} = \frac{1}{1925} = 0.000519.$$

The chi-square estimate of a is therefore $a = 0.770 \pm \sqrt{0.000519} = 0.770 \pm 0.023$.

Our fit has for χ^2 the value

$$\chi^2 = \frac{[0.8 - 1(0.770)]^2}{0.1^2} + \frac{[1.5 - 2(0.770)]^2}{0.05^2} + \frac{[2.7 - 3(0.770)]^2}{0.2^2} = 4.533.$$

Our problem involves three points and one parameter, and therefore its chi-square distribution has two degrees of freedom and, according to Eqs. (23.103) and (23.104), has a mean value of 2 and a variance of 4. Our value of χ^2 , 4.533, is significantly larger than the mean value of the distribution and therefore describes a data set with more spread than would normally be expected for the stated values of σ_j . We can obtain a more quantitative measure of the probability that χ^2 would be at least as large as our value by comparing with the entries in Table 23.2. Using the row of the table for $n = 2$, we see that the probability of getting a spread larger than that of our present data is quite small, only slightly above 0.1. ■

Student t Distribution

The Student t distribution (sometimes just called the t **distribution**) is designed for use with small data sets for which the variance is unknown. This distribution was first described by W. S. Gosset, who published his work under the pen name “Student” because his employer, the Guinness brewery, would not permit him to publish it under his own name.

Gosset considered the probability distribution of a random variable T , of the form

$$T = \frac{Y\sqrt{n}}{\sqrt{S/n}}. \quad (23.106)$$

For the applications under consideration here,

$$Y = \frac{1}{n} \sum_{j=1}^n X_j - \mu = \bar{X} - \mu, \quad (23.107)$$

$$S = \sum_{j=1}^n X_j^2. \quad (23.108)$$

Here the X_j are a set of n independent Gauss normal random variables, each of the same unknown variance σ^2 . The quantity μ is the (unknown) value of the mean of X . An important feature of Gosset’s choice for T is that (as we shall shortly show) its probability distribution $f_n(t)$ is independent of the variance of the X_j .

The procedure for obtaining the probability distribution of T depends on the fact that Y and S are independent random variables. That is so, but proof is beyond the scope of the present abbreviated discussion. We start by noting that S is a gamma distribution, with probability distribution $g(n, \sigma; s)$, as given in Eq. (23.85). Next, we proceed to the distribution of $U = \sqrt{S/n}$, which we denote $h(u)$. Making a change of variable from s to nu^2 , and observing that $ds = (ds/du)du$, we find

$$h(u) = g(n, \sigma; nu^2)(2nu). \quad (23.109)$$

This is the probability distribution of the denominator of T . To get the distribution of the numerator, we note that Y is a normal distribution with variance σ^2/n (see Eq. (23.92)), and mean zero. It, therefore (see Eq. (23.56)), has the distribution we denote $r(y)$, of the form

$$r(y) = \sqrt{\frac{n}{2\pi\sigma^2}} e^{-ny^2/2\sigma^2}. \quad (23.110)$$

The numerator, $Z = Y\sqrt{n}$, will therefore have a distribution $k(z)$, where $z = y\sqrt{n}$, so

$$k(z) = r(z/\sqrt{n})(dy/dz) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{z^2/2\sigma^2}; \quad (23.111)$$

the presence of the factor $\sqrt{n}$ causes the numerator to have variance σ^2 . Finally, we use the formula for the ratio of two independent distributions, Eq. (23.81), to obtain

$$f(t) = \int_0^\infty k(ut)h(u)u \, du. \quad (23.112)$$

The integration only extends from zero to infinity because the gamma distribution in $h(u)$ is only nonzero for positive u . Inserting expressions for the quantities in Eq. (23.112),

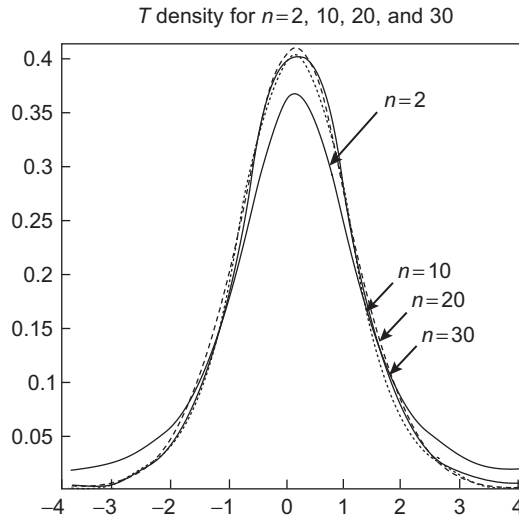
$$\begin{aligned} f_n(t) &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_0^\infty e^{-u^2t^2/2\sigma^2} g(n, \sigma; nu^2) (2nu^2) du \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} \frac{1}{2^{n/2}\sigma^n\Gamma(n/2)} \int_0^\infty e^{-u^2t^2/2\sigma^2} (nu^2)^{(n/2)-1} e^{-nu^2/2\sigma^2} (2nu^2) du \\ &= \frac{2}{\sigma^{n+1}\sqrt{\pi n}} \left(\frac{n}{2}\right)^{(n+1)/2} \frac{1}{\Gamma(n/2)} \int_0^\infty e^{-u^2(t^2+n)/2\sigma^2} u^n du. \end{aligned} \quad (23.113)$$

To complete the evaluation, we change variables in the integral to $z = u^2(t^2 + n)/2\sigma^2$, thereby making the integral identifiable as a gamma function, so

$$\begin{aligned} \int_0^\infty e^{-u^2(t^2+n)/2\sigma^2} u^n du &= \frac{1}{2} \left(\frac{2\sigma^2}{t^2+n}\right)^{(n+1)/2} \int_0^\infty z^{(n-1)/2} e^{-z} dz \\ &= \frac{1}{2} \left(\frac{2\sigma^2}{t^2+n}\right)^{(n+1)/2} \Gamma\left(\frac{n+1}{2}\right). \end{aligned} \quad (23.114)$$

Inserting this result into Eq. (23.113) and simplifying, we note that the instances of σ entirely cancel, and we are left with

$$f_n(t) = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{\pi n}\Gamma\left(\frac{n}{2}\right)} \left(1 + \frac{t^2}{n}\right)^{-(n+1)/2}. \quad (23.115)$$

FIGURE 23.13 Student t probability density $f_n(t)$ for $n = 2, 10, 20$, and 30 .

Equation (23.115) is the probability density for the t distribution with n degrees of freedom. This equation shows that we have achieved the desired result, namely that the distribution T is independent of the variance of the input random variables X_j . Since it is our intention to use the t distribution for the reduction of experimental data of unknown variance, we have achieved our current objective. Figure 23.13 shows densities $f_n(t)$ for several n ; an important feature of these curves is that they depend very weakly on n .

Confidence Intervals

A **confidence interval** for a random variable X is the range within which x will fall, not with certainty but with a high probability, the confidence level, which we can choose. If X has a probability distribution $f(x)$, the confidence interval for probability p will be the range of x , usually symmetrically centered about some value x_0 , that contains the fraction p of its probability distribution. If this range is bounded by $x_0 - dx$ and $x_0 + dx$, it is customary to write that $x = x_0 \pm dx$ with $(100p)\%$ confidence. If, for example, a computed value of x is 0.50 and 90% of its probability distribution falls between $x = 0.40$ and $x = 0.60$, we say that x has the value $x = 0.50 \pm 0.10$ with 90% confidence.

Confidence intervals are usually found by what is called the **pivotal method**, which involves relating the variable for which we desire a confidence interval to a known probability distribution. The identification and selection of pivotal quantities is in general outside the scope of this text, but for a Gauss normal random variable with zero mean, a suitable pivot is its t distribution. Referring to Eq. (23.106), this means we can estimate a confidence interval for Y (the deviations of the observed mean $\bar{X}$ from its true value) from the equation

$$Y = \bar{X} - \mu = \frac{T\sqrt{S/n}}{\sqrt{n}}, \quad (23.116)$$

where T is the random variable corresponding to the t distribution and S is the single value obtained by inserting the observed values of the X_i into Eq. (23.108). We use Eq. (23.116) by inserting into it the range of T that corresponds to a total probability p , calculating therefrom the corresponding range of Y . Note that we do not insert a probability distribution for S ; we use the value of S arising from our data.

The distribution of T is an even function of t with a maximum at $t = 0$, as is obvious from Eq. (23.115) and Fig. 23.13, and our confidence interval for T will naturally be centered about zero. Therefore, a confidence interval of probability p will correspond to a symmetric range of t , $(-C_p < t < +C_p)$, such that

$$P(-C_p < t < +C_p) = p.$$

Because $f(t)$ is even, we also have

$$P(-\infty < t < +C_p) = \frac{1}{2} + (-C_p < t < C_p)/2,$$

which is equivalent to

$$P(-\infty < t < +C_p) = \frac{1+p}{2} \equiv \hat{p}. \quad (23.117)$$

Because of the frequent need to use values of C corresponding to various values of $\hat{p}$ and degrees of freedom n , these C values have been tabulated and appear in many statistics texts. A short table is included here (Table 23.3).

Given a confidence interval for T , we may insert it into Eq. (23.116), which when solved for μ becomes

$$\mu = \bar{X} - T \frac{\sqrt{S/n}}{\sqrt{n}} = \bar{x} - \frac{T\sigma}{\sqrt{n}}. \quad (23.118)$$

From the limiting values for T , we get the corresponding range for μ , which is valid with the probability of the T range. Note that except for the range of T , all the quantities on the right-hand side of Eq. (23.118) are to be computed from our sample data. In particular, we need the mean value $\bar{X}$ for our sample and the standard deviation of our data points, $\sigma = \sqrt{S/n}$. Note further that, as with the chi-square distribution, when measured data are used to generate the sample mean and sample standard deviation, the appropriate t distribution to use for n data points is that with $n - 1$ degrees of freedom, and in using

Table 23.3 Student t Distribution

$\hat{p}$	$n = 1$	$n = 2$	$n = 3$	$n = 4$	$n = 5$
0.8	1.38	1.06	0.98	0.94	0.92
0.9	3.08	1.89	1.64	1.53	1.48
0.95	6.31	2.92	2.35	2.13	2.02
0.975	12.7	4.30	3.18	2.78	2.57
0.99	31.8	6.96	4.54	3.75	3.36
0.999	318.3	22.3	10.2	7.17	5.89

Note: Entries are the values C in $\int_{-\infty}^C f_n(t) dt = \hat{p}$, where $f_n(t)$ is given in Eq. (23.115), with n the number of degrees of freedom.

Eq. (23.118) it is customary to take σ as the sample standard deviation, as defined in Eq. (23.95). These points are explained more fully in several of the Additional Readings.

Example 23.7.3 CONFIDENCE INTERVAL

Suppose we have the following random data from a population that can be assumed to have a Gauss normal distribution:

$$7.12 \quad 4.95 \quad 6.18 \quad 5.69 \quad 2.90 \quad 8.47,$$

and we wish to determine 90% and 95% confidence intervals for the population mean.

Since we have neither the population mean nor variance, but have assumed the population distribution to be normal, we can use the t distribution as just outlined. As a preliminary to doing so, we need to calculate the sample mean and standard deviation. Since we have six data points, the number of degrees of freedom will be $n = 5$. We have

$$\begin{aligned}\bar{X} &= (7.12 + 4.95 + 6.18 + 5.60 + 2.90 + 8.47)/6 = 5.885, \\ \sigma &= \left[\frac{1}{5} \left((7.12 - 5.885)^2 + \cdots + (8.47 - 5.885)^2 \right) \right]^{1/2} = 1.9035.\end{aligned}$$

Considering first the 90% confidence interval that corresponds to the range $(-C_{90} < t < C_{90})$ with $\hat{p} = (1 + p)/2 = 0.95$, we read from Table 23.3 the value $C_{90} = 2.02$. Thus,

$$\mu = 5.885 \pm \frac{(2.02)(1.9035)}{\sqrt{5}} = 5.885 \pm 1.720 \quad (90\% \text{ confidence}).$$

For 95% confidence, we need C_{95} , again for $n = 5$. This time, $\hat{p} = 0.975$, so $C_{95} = 2.57$, and

$$\mu = 5.885 \pm \frac{(2.57)(1.9035)}{\sqrt{5}} = 5.885 \pm 2.188 \quad (95\% \text{ confidence}).$$

A few final observations are in order. First, we see that by demanding an increase in the confidence level, the interval probably containing the true mean becomes wider. Note that at high confidence levels the probable width can become much larger than the sample standard deviation. Finally, note that even the confidence intervals are sample-dependent. Other data from the same population could generate intervals of different widths. Perhaps oversimplifying, these analyses show that there is no way of converting probability data into significant statements that have complete certainty. ■

Exercises

23.7.1 Let ΔA be the error of a measurement of A , etc. Use error propagation to show that

$$\left(\frac{\sigma(C)}{C} \right)^2 = \left(\frac{\sigma(A)}{A} \right)^2 + \left(\frac{\sigma(B)}{B} \right)^2$$

holds for the product $C = AB$ and the ratio $C = A/B$.

- 23.7.2** Find the mean value and standard deviation of the sample of measurements $x_1 = 6.0$, $x_2 = 6.5$, $x_3 = 5.9$, $x_4 = 6.1$, $x_5 = 6.2$. If the point $x_6 = 6.1$ is added to the sample, how does the change affect the mean value and standard deviation?
- 23.7.3** Carry out a χ^2 analysis of the fit corresponding to Fig. 23.10 using the same points as in Example 23.7.2, but with the errors now associated with the t_i rather than the y_i .
- 23.7.4** Using the data from Exercise 23.7.2 (including the point x_6), find the 90% and 95% confidence intervals for the mean of the x_i .

Additional Readings

- Bevington, P. R., and D. K. Robinson, *Data Reduction and Error Analysis for the Physical Sciences*, 3rd ed. New York: McGraw-Hill (2003).
- Chung, K. L., *A Course in Probability Theory Revised*, 3rd ed. New York: Academic Press (2000).
- DeGroot, M. H., *Probability and Statistics*, 2nd ed. Reading, MA: Addison-Wesley (1986).
- Devore, J. L., *Probability and Statistics for Engineering and the Sciences*, 5th ed. New York: Duxbury Press (1999).
- Freund, J. E., and R. E. Walpole, *Mathematical Statistics*, 4th ed. Englewood Cliffs, NJ: Prentice Hall (1987). This well-regarded text is at a level comparable to the exposition in this chapter. Clear and with many statistical tables.
- Kreyszig, E., *Introductory Mathematical Statistics: Principles and Methods*. New York: Wiley (1970).
- Montgomery, D. C., and G. C. Runger, *Applied Statistics and Probability for Engineers*, 2nd ed. New York: Wiley (1998).
- Papoulis, A., *Probability, Random Variables, and Stochastic Processes*, 3rd ed. New York: McGraw-Hill (1991).
- Ramachandran, K. M., and C. P. Tsokos, *Mathematical Statistics with Applications*. New York: Academic Press (2009). Relatively detailed but readable and self-contained.
- Ross, S. M., *First Course in Probability*, 5th ed., vol. A. New York: Prentice Hall (1997).
- Ross, S. M., *Introduction to Probability and Statistics for Engineers and Scientists*, 2nd ed. New York: Academic Press (1999).
- Ross, S. M., *Introduction to Probability Models*, 7th ed. New York: Academic Press (2000).
- Suhir, E., *Applied Probability for Engineers and Scientists*. New York: McGraw-Hill (1997).

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